



MISO DPP 2017 August West Area Study Phase 1 Final Report

October 3, 2019

**MISO
720 City Center
Drive Carmel
Indiana 46032**
<http://www.misoenergy.org>

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1. Executive Summary

This report presents the results of a System Impact Study (SIS) performed to evaluate the interconnection of the generators in the DPP 2017 August West Area Phase 1 (West Area DPP 1). The study was performed under the direction of MISO and reviewed by an ad hoc study group. The ad hoc study group was formed to review the study scope, methodology, models and results. The ad hoc study group consisted of representatives from the interconnection customers and the following utility companies – American Transmission Company, Ameren, Basin Electric Power Cooperative, CIPCO, Cedar Falls, Dairyland Power Cooperative, Great River Energy, ITC Midwest, Montana-Dakota Utilities, MidAmerican, Minnesota Power, Ottertail Power, SMMPA, and Xcel Energy.

1.1. Project List

The original interconnection requests for DPP 2017 August West Area had a total of 36 projects generation projects. During the model review, two projects withdrew. Therefore, there are 34 projects generation projects with a combined nameplate rating of 4981.76 MW (ERIS) & 4319.26 MW (NRIS). The detailed list of West Area DPP 1 is shown below in Table 1, and the Phase 1 study was kicked off on June 12, 2019.

Table 1: List of DPP August 2017 West Area Phase 1 Projects

Project	Fuel Type	Transmission Owner	County	State	Service Requested	MW	POI
G517	Wind	ITCM	Cottonwood	MN	ERIS	130	Summit Switch Station South Storden 69 kV
J447	Wind	ITCM	Cerro Gordo	IA	NRIS	150	Emery 161 kV
J545	Wind	Xcel	Lincoln	MN	NRIS	110	Buffalo Ridge 115 kV
J580	Wind	MDU	Burleigh	ND	ERIS	298	Wishek-Hesket 230 kV
J628	Wind	GRE	Grand Forks, Nelson	ND	NRIS	400	Prairie-Ramsey 230 kV
J705	Wind	MP	Morton	ND	NRIS	100	Tri-County 230 kV
J706	Wind	MP	Morton	ND	NRIS	100	Tri-County 230 kV
J713	Wind	MP	Oliver	ND	NRIS	300	Square-Butte 230 kV
J720	Wind	ITCM	Jackson	MN	NRIS	200	Lakefield-Raun 345 kV
J722	Wind	OTP	Codington, Deuel	SD	NRIS	200	Big Stone South 230 kV
J743	Wind	Xcel	Cass	ND	NRIS	200	Bison 345 kV
J785	Wind	ITCM	Jackson, Martin	MN	NRIS	105	Lakefield Junction-Huntley 345 kV
J801	Solar	DPC	Pierce	WI	NRIS	74	Crystal Cave – Rock Elm 161 kV
J803	Solar	Xcel	Lyon	MN	ERIS	32.5	Tracy-Walnut Grove 69 kV
J816	Solar	OTP	Cass	ND	NRIS	60	Buffalo 115 kV
J836	Wind	ITCM	Kossuth	IA	ERIS	200	Ledyard 345 kV
J840	Wind	MEC	Humboldt	IA	NRIS	150	Kossuth-Webster 345 kV
J869	Solar	GRE	Sherburne	MN	NRIS	49.98	Cable-Cable Tap 69 kV
J873	Wind	ITCM	Hardin	IA	NRIS	200	Emery-Blackhawk 345 kV
J874	Solar	Xcel	Murray	MN	NRIS	150	Fenton-Chanarambie 115 kV
J876*	Wind	ITCM	Faribault	MN	NRIS	200	Huntley 345 kV
J877	Solar	MEC	Palo Alto	IA	NRIS	250	Palo Alto 345 kV
J880	Wind	Xcel	Ward	ND	NRIS	150	Magic City 230 kV
J885	Wind	ITCM	Freeborn,	IA,	NRIS	64	Glenworth 161 kV

Project	Fuel Type	Transmission Owner	County	State	Service Requested	MW	POI
			Worth	MN			
J889	Wind	GRE	Nelson	ND	NRIS	150	Ramsey-Prairie 230 kV
J891*	Wind	Xcel	Pipestone	MN	NRIS	300	Buffalo Ridge-Pipestone 115 kV
J897	Wind	GRE	Grand Forks	ND	NRIS	190	Ramsey-Prairie 230 kV
J898	Wind	DPC	Fillmore, Mower	MN	NRIS	100	Beaver Creek 161 kV
J901	Wind	Xcel	Redwood	MN	NRIS	200	Lyon County-Cedar Mountain 345 kV
J905	Solar	Xcel	Pipestone	MN	NRIS	40	Buffalo Ridge 115 kV
J916	Diesel	ITCM	Cottonwood	MN	E-NRIS	2	Mountain Lake 69 kV
J926	Wind	Xcel	Saint Croix	WI	NRIS	101.28	Pine Lake – Apple River 161 kV
J927	Wind	ITCM	Faribault	MN	NRIS	100	Walters 69 kV Substation
J929	Solar	MDU	McIntosh	ND	NRIS	25	Wishek 41.6 kV
J933	Wind	OTP	Day	SD	NRIS	200	Ellendale-Big Stone South 345 kV
J946	Solar	Xcel	Cass	ND	NRIS	200	Bison 345 kV

*Denotes projects withdrawn after Kick-off



1.2. Project Summary Network Upgrades

G517			
Network Upgrade	NU Cost Estimate (\$)	G517 Cost Estimate (\$)	NU Type
Cottonwood-Highway Tap Rebuild	\$725,000	\$725,000	ERIS Thermal
Cottonwood-Waterbury Rebuild	\$4,400,000	\$4,400,000	ERIS Thermal
Highway Tap-Lambrtn Rebuild	\$334,000	\$334,000	ERIS Thermal
Jefferson Tap-N Storden Rebuild	\$2,900,000	\$2,900,000	ERIS Thermal
Jefferson Tap-Storden Rebuild	\$900,000	\$900,000	ERIS Thermal
Lambrtn-N Storden Rebuild	\$5,900,000	\$5,900,000	ERIS Thermal
S Storden-Heron Lake Rebuild	\$8,400,000	\$8,400,000	ERIS Thermal
S Storden-Storden Rebuild	\$4,000,000	\$4,000,000	ERIS Thermal
Heron Lake Transformer 1 Upgrade	\$2,520,000	\$2,520,000	ERIS Thermal
Heron Lake Transformer 2 Upgrade	\$2,520,000	\$2,520,000	ERIS Thermal
Windms-Wilder Uprate	\$100,000	\$100,000	ERIS Thermal
Wanda-Sundown 69 kV Rebuild	\$3,150,000	\$3,150,000	ERIS Thermal
Wanda-Waterbury 69 kV Rebuild	\$1,500,000	\$1,500,000	ERIS Thermal
SpringM-DOTCORN 69 kV Uprate	\$60,000	\$60,000	ERIS Thermal
Brookville-Sundown 69 kV Rebuild	\$2,700,000	\$2,700,000	ERIS Thermal
Brookville-COBDN Tap 69 kV Rebuild	\$2,570,000	\$2,570,000	ERIS Thermal
SpringM-COBDN Tap 69 kV Rebuild	\$396,000	\$396,000	ERIS Thermal
Twin Brooks-Alexandria	\$242,400,000	\$1,989,062	ERIS Voltage
Alexandria 4 steps of 50 MVar Capacitor banks and 345 kV bus expansion	\$26,000,000	\$408,043	ERIS Voltage
Additional Bison Breaker	\$1,500,000	\$1,138	OTP LPC
Silver Lake 60 MVar Capacitor Bank	\$2,500,000	\$23,199	GRE LPC
Wahpeton 3 steps of 25 MVar Capacitor Banks	\$3,000,000	\$17,834	OTP LPC
Audubon 40 MVar Capacitor Bank	\$1,500,000	\$2,780	OTP LPC
Bagley 20 MVar Capacitor Bank	\$1,000,000	\$2,132	OTP LPC
Alexandria 115 kV 16.2 MVar Capacitor Bank (OTP Owned)	\$1,000,000	\$5,491	OTP LPC
Panther 60 MVar Capacitor Bank	\$2,500,000	\$51,566	GRE LPC
GRE Prairie Voltage Collapse Network Upgrade	\$211,350,000	\$43,486	GRE LPC
Total Cost:		\$45,619,731	



J447			
Network Upgrade	NU Cost Estimate (\$)	J447 Cost Estimate (\$)	NU Type
Twin Brooks-Alexandria	\$242,400,000	\$497,265	ERIS Voltage
Alexandria 4 steps of 50 MVar Capacitor banks and 345 kV bus expansion	\$26,000,000	\$25,503	ERIS Voltage
Total Cost Per Project:		\$522,768	

J545			
Network Upgrade	NU Cost Estimate (\$)	J545 Cost Estimate (\$)	NU Type
Twin Brooks-Alexandria	\$242,400,000	\$3,480,858	ERIS Voltage
Alexandria 4 steps of 50 MVar Capacitor banks and 345 kV bus expansion	\$26,000,000	\$803,335	ERIS Voltage
Additional Bison Breaker	\$1,500,000	\$8,747	OTP LPC
Silver Lake 60 MVar Capacitor Bank	\$2,500,000	\$66,141	GRE LPC
Wahpeton 3 steps of 25 MVar Capacitor Banks	\$3,000,000	\$53,161	OTP LPC
Audubon 40 MVar Capacitor Bank	\$1,500,000	\$10,994	OTP LPC
Bagley 20 MVar Capacitor Bank	\$1,000,000	\$5,947	OTP LPC
Alexandria 115 kV 16.2 MVar Capacitor Bank (OTP Owned)	\$1,000,000	\$18,517	OTP LPC
Panther 60 MVar Capacitor Bank	\$2,500,000	\$146,234	GRE LPC
GRE Prairie Voltage Collapse Network Upgrade	\$211,350,000	\$211,958	GRE LPC
CCS GRE LPC Voltage Network Upgrade	\$26,500,000	\$20,366	GRE LPC
Total Cost Per Project:		\$4,826,258	

J580			
Network Upgrade	NU Cost Estimate (\$)	J580 Cost Estimate (\$)	NU Type
New Mandon-J580 230 kV Line	\$28,000,000	\$27,544,485	ERIS Thermal/ERIS Voltage
Rebuild Merricourt - Wishek 230 kV	\$5,800,000	\$5,330,419	ERIS Thermal
Merricourt - Tatanka Structure Remediation	\$100,000	\$91,670	ERIS Thermal
Sheyenne-Lake Park Uprate	\$1,300,000	\$317,957	ERIS Thermal
Twin Brooks-Alexandria	\$242,400,000	\$24,864,507	ERIS Voltage
Alexandria 4 steps of 50 MVar Capacitor banks and 345 kV bus expansion	\$26,000,000	\$2,205,983	ERIS Voltage
Additional Merricourt Breakers	\$1,500,000	\$1,233,890	MDU LPC
Foxtail-Tatanka North 230 kV Upgrade	\$2,000,000	\$1,663,404	MDU LPC
Foxtail-Ellendale 230 kV Rebuild	\$10,500,000	\$8,732,869	MDU LPC
Merricourt-Tatanka North 230 kV Upgrade	\$1,000,000	\$831,702	MDU LPC
Merricourt-Ellendale 230 kV Reconnector	\$18,000,000	\$14,970,560	MDU LPC
Additional Bison Breaker	\$1,500,000	\$190,952	OTP LPC
Silver Lake 60 MVar Capacitor Bank	\$2,500,000	\$330,292	GRE LPC
Jamestown 75 MVar Capacitor Bank	\$4,000,000	\$710,630	OTP LPC



Wahpeton 3 steps of 25 MVar Capacitor Banks	\$3,000,000	\$455,076	OTP LPC
Audubon 40 MVar Capacitor Bank	\$1,500,000	\$184,926	OTP LPC
Perham 2 steps of 20 MVar Capacitor Banks	\$2,000,000	\$175,894	OTP LPC
Bagley 20 MVar Capacitor Bank	\$1,000,000	\$90,145	OTP LPC
Alexandria 115 kV 16.2 MVar Capacitor Bank (OTP Owned)	\$1,000,000	\$104,214	OTP LPC
Panther 60 MVar Capacitor Bank	\$2,500,000	\$233,441	GRE LPC
GRE Prairie Voltage Collapse Network Upgrade	\$211,350,000	\$10,096,519	GRE LPC
CCS GRE LPC Voltage Network Upgrade	\$26,500,000	\$5,908,334	GRE LPC
Total Cost Per Project:		\$106,267,869	

J628			
Network Upgrade	NU Cost Estimate (\$)	J628 Cost Estimate (\$)	NU Type
Mahnommn-Ulrich Uprate	\$50,000	\$50,000	ERIS Thermal
Winger-Fertile Uprate	\$100,000	\$100,000	ERIS Thermal
Wilton-Bemidji Uprate	\$50,000	\$28,210	ERIS Thermal
Crookston-Fertile Uprate	\$250,000	\$250,000	ERIS Thermal
Crookston-Falconer Uprate	\$25,000	\$25,000	ERIS Thermal
Riverton-Mud Lake Uprate	\$882,000	\$882,000	ERIS Thermal
Twin Brooks-Alexandria	\$242,400,000	\$27,064,951	ERIS Voltage
Rebuild J628-Prairie 230 kV Rebuild	\$16,770,000	\$9,732,051	ERIS Thermal
Alexandria 4 steps of 50 MVar Capacitor banks and 345 kV bus expansion	\$26,000,000	\$2,461,010	ERIS Voltage
New J628-Prairie 230 kV 2nd Circuit	\$22,360,000	\$8,653,870	ERIS Voltage
Winger-Fertile 115 kV Uprate (OTP LPC)	\$450,000	\$450,000	OTP LPC
Crookston-Fertile 115 kV Uprate (OTP LPC)	\$250,000	\$250,000	OTP LPC
Donalds-Warsaw 115 kV Uprate	\$50,000	\$50,000	OTP LPC
Wilton-Bemidji 115 kV Uprate (OTP LPC)	\$50,000	\$34,615	OTP LPC
Additional Bison Breaker	\$1,500,000	\$233,406	OTP LPC
Silver Lake 60 MVar Capacitor Bank	\$2,500,000	\$209,575	GRE LPC
Jamestown 75 MVar Capacitor Bank	\$4,000,000	\$343,038	OTP LPC
Wahpeton 3 steps of 25 MVar Capacitor Banks	\$3,000,000	\$238,524	OTP LPC
Audubon 40 MVar Capacitor Bank	\$1,500,000	\$174,512	OTP LPC
Perham 2 steps of 20 MVar Capacitor Banks	\$2,000,000	\$564,759	OTP LPC
Bagley 20 MVar Capacitor Bank	\$1,000,000	\$283,306	OTP LPC
Alexandria 115 kV 16.2 MVar Capacitor Bank (OTP Owned)	\$1,000,000	\$81,993	OTP LPC



Panther 60 MVar Capacitor Bank	\$2,500,000	\$194,683	GRE LPC
GRE Prairie Voltage Collapse Network Upgrade	\$211,350,000	\$94,419,919	GRE LPC
Total Cost Per Project:		\$146,775,422	

J705			
Network Upgrade	NU Cost Estimate (\$)	J705 Cost Estimate (\$)	NU Type
Twin Brooks-Alexandria	\$242,400,000	\$8,811,007	ERIS Voltage
Alexandria 4 steps of 50 MVar Capacitor banks and 345 kV bus expansion	\$26,000,000	\$1,313,389	ERIS Voltage
Additional Bison Breaker	\$1,500,000	\$70,623	OTP LPC
Silver Lake 60 MVar Capacitor Bank	\$2,500,000	\$92,720	GRE LPC
Jamestown 75 MVar Capacitor Bank	\$4,000,000	\$427,505	OTP LPC
Wahpeton 3 steps of 25 MVar Capacitor Banks	\$3,000,000	\$121,917	OTP LPC
Audubon 40 MVar Capacitor Bank	\$1,500,000	\$70,047	OTP LPC
Perham 2 steps of 20 MVar Capacitor Banks	\$2,000,000	\$78,979	OTP LPC
Bagley 20 MVar Capacitor Bank	\$1,000,000	\$36,081	OTP LPC
Alexandria 115 kV 16.2 MVar Capacitor Bank (OTP Owned)	\$1,000,000	\$34,816	OTP LPC
Panther 60 MVar Capacitor Bank	\$2,500,000	\$70,777	GRE LPC
GRE Prairie Voltage Collapse Network Upgrade	\$211,350,000	\$5,601,061	GRE LPC
CCS GRE LPC Voltage Network Upgrade	\$26,500,000	\$3,510,118	GRE LPC
Total Cost Per Project:		\$20,239,040	

J706			
Network Upgrade	NU Cost Estimate (\$)	J705 Cost Estimate (\$)	NU Type
Twin Brooks-Alexandria	\$242,400,000	\$8,811,007	ERIS Voltage
Alexandria 4 steps of 50 MVar Capacitor banks and 345 kV bus expansion	\$26,000,000	\$1,313,389	ERIS Voltage
Additional Bison Breaker	\$1,500,000	\$70,623	OTP LPC
Silver Lake 60 MVar Capacitor Bank	\$2,500,000	\$92,720	GRE LPC
Jamestown 75 MVar Capacitor Bank	\$4,000,000	\$427,505	OTP LPC
Wahpeton 3 steps of 25 MVar Capacitor Banks	\$3,000,000	\$121,917	OTP LPC
Audubon 40 MVar Capacitor Bank	\$1,500,000	\$70,047	OTP LPC
Perham 2 steps of 20 MVar Capacitor Banks	\$2,000,000	\$78,979	OTP LPC



Bagley 20 MVar Capacitor Bank	\$1,000,000	\$36,081	OTP LPC
Alexandria 115 kV 16.2 MVar Capacitor Bank (OTP Owned)	\$1,000,000	\$34,816	OTP LPC
Panther 60 MVar Capacitor Bank	\$2,500,000	\$70,777	GRE LPC
GRE Prairie Voltage Collapse Network Upgrade	\$211,350,000	\$5,601,061	GRE LPC
CCS GRE LPC Voltage Network Upgrade	\$26,500,000	\$3,510,118	GRE LPC
Total Cost Per Project:		\$20,239,040	

J713			
Network Upgrade	NU Cost Estimate (\$)	J713 Cost Estimate (\$)	NU Type
Sheyenne-Lake Park Uprate	\$1,300,000	\$362,229	ERIS Thermal
Twin Brooks-Alexandria	\$242,400,000	\$24,057,901	ERIS Voltage
Alexandria 4 steps of 50 MVar Capacitor banks and 345 kV bus expansion	\$26,000,000	\$2,333,497	ERIS Voltage
Additional Bison Breaker	\$1,500,000	\$211,869	OTP LPC
Silver Lake 60 MVar Capacitor Bank	\$2,500,000	\$278,159	GRE LPC
Jamestown 75 MVar Capacitor Bank	\$4,000,000	\$1,282,515	OTP LPC
Wahpeton 3 steps of 25 MVar Capacitor Banks	\$3,000,000	\$365,752	OTP LPC
Audubon 40 MVar Capacitor Bank	\$1,500,000	\$210,141	OTP LPC
Perham 2 steps of 20 MVar Capacitor Banks	\$2,000,000	\$236,937	OTP LPC
Bagley 20 MVar Capacitor Bank	\$1,000,000	\$108,244	OTP LPC
Alexandria 115 kV 16.2 MVar Capacitor Bank (OTP Owned)	\$1,000,000	\$104,448	OTP LPC
Panther 60 MVar Capacitor Bank	\$2,500,000	\$212,332	GRE LPC
GRE Prairie Voltage Collapse Network Upgrade	\$211,350,000	\$16,803,182	GRE LPC
CCS GRE LPC Voltage Network Upgrade	\$26,500,000	\$10,530,353	GRE LPC
Total Cost Per Project:		\$57,097,559	

J720			
Network Upgrade	NU Cost Estimate (\$)	J720 Cost Estimate (\$)	NU Type
Twin Brooks-Alexandria	\$242,400,000	\$1,989,062	ERIS Voltage
Alexandria 4 steps of 50 MVar Capacitor banks and 345 kV bus expansion	\$26,000,000	\$293,281	ERIS Voltage
Silver Lake 60 MVar Capacitor Bank	\$2,500,000	\$14,144	GRE LPC
Wahpeton 3 steps of 25 MVar Capacitor Banks	\$3,000,000	\$16,890	OTP LPC
Audubon 40 MVar Capacitor Bank	\$1,500,000	\$670	OTP LPC
Bagley 20 MVar Capacitor Bank	\$1,000,000	\$1,640	OTP LPC



Alexandria 115 kV 16.2 MVar Capacitor Bank (OTP Owned)	\$1,000,000	\$4,659	OTP LPC
Panther 60 MVar Capacitor Bank	\$2,500,000	\$14,359	GRE LPC
GRE Prairie Voltage Collapse Network Upgrade	\$211,350,000	\$17,074	GRE LPC
Total Cost Per Project:		\$2,351,779	

J722			
Network Upgrade	NU Cost Estimate (\$)	J722 Cost Estimate (\$)	NU Type
Twin Brooks-Alexandria	\$242,400,000	\$11,934,369	ERIS Voltage
Alexandria 4 steps of 50 MVar Capacitor banks and 345 kV bus expansion	\$26,000,000	\$2,078,470	ERIS Voltage
Canby-Granite Falls 115 kV Rebuild	\$17,700,000	\$9,654,545	OTP LPC
Additional Bison Breaker	\$1,500,000	\$31,908	OTP LPC
Silver Lake 60 MVar Capacitor Bank	\$2,500,000	\$267,656	GRE LPC
Wahpeton 3 steps of 25 MVar Capacitor Banks	\$3,000,000	\$330,866	OTP LPC
Audubon 40 MVar Capacitor Bank	\$1,500,000	\$39,607	OTP LPC
Bagley 20 MVar Capacitor Bank	\$1,000,000	\$22,050	OTP LPC
Alexandria 115 kV 16.2 MVar Capacitor Bank (OTP Owned)	\$1,000,000	\$85,999	OTP LPC
Panther 60 MVar Capacitor Bank	\$2,500,000	\$275,611	GRE LPC
GRE Prairie Voltage Collapse Network Upgrade	\$211,350,000	\$478,762	GRE LPC
Total Cost Per Project:		\$25,199,843	

J743			
Network Upgrade	NU Cost Estimate (\$)	J743 Cost Estimate (\$)	NU Type
Sheyenne-Lake Park Uprate	\$1,300,000	\$463,050	ERIS Thermal
Sheyenne-Mapleton Rebuild	\$17,600,000	\$10,782,897	ERIS Thermal
Granite City-Cross Roads Rebuild	\$3,700,000	\$3,700,000	ERIS Thermal
Cross Roads-MEI International Rebuild	\$1,300,000	\$1,300,000	ERIS Thermal
MEI International-Westwood Rebuild	\$2,000,000	\$2,000,000	ERIS Thermal
Westwood-West St. Cloud Rebuild	\$600,000	\$600,000	ERIS Thermal
Buffalo Transformer #1 Upgrade	\$3,000,000	\$1,997,532	ERIS Thermal
Buffalo Transformer #2 Upgrade	\$3,000,000	\$1,997,532	ERIS Thermal
Mapleton-CSLNET Uprate	\$25,000	\$14,382	ERIS Thermal
Badura-Hubbard Uprate	\$600,000	\$415,949	ERIS Thermal
Hubbard-Audubon Uprate	\$2,000,000	\$2,000,000	ERIS Thermal
Twin Brooks-Alexandria	\$242,400,000	\$16,907,023	ERIS Voltage



Alexandria 4 steps of 50 MVar Capacitor banks and 345 kV bus expansion	\$26,000,000	\$2,461,010	ERIS Voltage
Wahpeton-Fergus Falls 230 kV Uprate	\$1,150,000	\$388,716	OTP LPC
Audubon-Hubbard 230 kV Uprate (OTP LPC)	\$250,000	\$166,667	OTP LPC
Additional Bison Breaker	\$1,500,000	\$207,387	OTP LPC
Silver Lake 60 MVar Capacitor Bank	\$2,500,000	\$233,650	GRE LPC
Wahpeton 3 steps of 25 MVar Capacitor Banks	\$3,000,000	\$313,828	OTP LPC
Audubon 40 MVar Capacitor Bank	\$1,500,000	\$256,392	OTP LPC
Perham 2 steps of 20 MVar Capacitor Banks	\$2,000,000	\$180,931	OTP LPC
Bagley 20 MVar Capacitor Bank	\$1,000,000	\$62,019	OTP LPC
Alexandria 115 kV 16.2 MVar Capacitor Bank (OTP Owned)	\$1,000,000	\$130,630	OTP LPC
Panther 60 MVar Capacitor Bank	\$2,500,000	\$123,726	GRE LPC
GRE Prairie Voltage Collapse Network Upgrade	\$211,350,000	\$298,268	GRE LPC
Total Cost Per Project:		\$47,001,589	

J785			
Network Upgrade	NU Cost Estimate (\$)	J785 Cost Estimate (\$)	NU Type
Twin Brooks-Alexandria	242,400,000	\$994,531	ERIS Voltage
Alexandria 4 steps of 50 MVar Capacitor banks and 345 kV bus expansion	\$26,000,000	\$165,768	ERIS Voltage
Silver Lake 60 MVar Capacitor Bank	\$2,500,000	\$4,835	GRE LPC
Wahpeton 3 steps of 25 MVar Capacitor Banks	\$3,000,000	\$6,428	OTP LPC
Bagley 20 MVar Capacitor Bank	\$1,000,000	\$542	OTP LPC
Alexandria 115 kV 16.2 MVar Capacitor Bank (OTP Owned)	\$1,000,000	\$1,663	OTP LPC
Panther 60 MVar Capacitor Bank	\$2,500,000	\$2,366	GRE LPC
Total Cost Per Project:		\$1,176,133	

J801			
Network Upgrade	NU Cost Estimate (\$)	J801 Cost Estimate (\$)	NU Type
	\$0	\$0	
Total Cost Per Project:		\$0	



J803			
Network Upgrade	NU Cost Estimate (\$)	J803 Cost Estimate (\$)	NU Type
Twin Brooks-Alexandria	\$242,400,000	\$497,265	ERIS Voltage
Alexandria 4 steps of 50 MVar Capacitor banks and 345 kV bus expansion	\$26,000,000	\$165,768	ERIS Voltage
Additional Bison Breaker	\$1,500,000	\$861	OTP LPC
Silver Lake 60 MVar Capacitor Bank	\$2,500,000	\$8,025	GRE LPC
Wahpeton 3 steps of 25 MVar Capacitor Banks	\$3,000,000	\$4,464	OTP LPC
Audubon 40 MVar Capacitor Bank	\$1,500,000	\$1,117	OTP LPC
Bagley 20 MVar Capacitor Bank	\$1,000,000	\$612	OTP LPC
Alexandria 115 kV 16.2 MVar Capacitor Bank (OTP Owned)	\$1,000,000	\$1,544	OTP LPC
Panther 60 MVar Capacitor Bank	\$2,500,000	\$22,730	GRE LPC
GRE Prairie Voltage Collapse Network Upgrade	\$211,350,000	\$10,985	GRE LPC
Total Cost Per Project:		\$713,371	

J816			
Network Upgrade	NU Cost Estimate (\$)	J816 Cost Estimate (\$)	NU Type
Sheyenne-Lake Park Uprate	\$1,300,000	\$60,774	ERIS Thermal
Sheyenne-Mapleton Rebuild	\$17,600,000	\$3,306,998	ERIS Thermal
Mapleton-CSLTNET Uprate	\$25,000	\$3,400	ERIS Thermal
Badura-Hubbard Uprate	\$600,000	\$43,822	ERIS Thermal
Twin Brooks-Alexandria	\$242,400,000	\$2,486,327	ERIS Voltage
Alexandria 4 steps of 50 MVar Capacitor banks and 345 kV bus expansion	\$26,000,000	\$293,281	ERIS Voltage
Addition Bison Breaker	\$1,500,000	\$33,916	OTP LPC
Silver Lake 60 MVar Capacitor Bank	\$2,500,000	\$35,020	GRE LPC
Wahpeton 3 steps of 25 MVar Capacitor Banks	\$3,000,000	\$47,019	OTP LPC
Audubon 40 MVar Capacitor Bank	\$1,500,000	\$38,810	OTP LPC
Perham 2 steps of 20 MVar Capacitor Banks	\$2,000,000	\$28,875	OTP LPC
Bagley 20 MVar Capacitor Bank	\$1,000,000	\$9,576	OTP LPC
Alexandria 115 kV 16.2 MVar Capacitor Bank (OTP Owned)	\$1,000,000	\$17,973	OTP LPC
Panther 60 MVar Capacitor Bank	\$2,500,000	\$18,990	GRE LPC
GRE Prairie Voltage Collapse Network Upgrade	\$211,350,000	\$188,316	GRE LPC
Total Cost Per Project:		\$6,613,097	

J836			
Network Upgrade	NU Cost Estimate (\$)	J836 Cost Estimate (\$)	NU Type
Twin Brooks-Alexandria	\$242,400,000	\$1,491,796	ERIS Voltage
Alexandria 4 steps of 50 MVar Capacitor banks and 345 kV bus expansion	\$26,000,000	\$165,768	ERIS Voltage
Silver Lake 60 MVar Capacitor Bank	\$2,500,000	\$241	GRE LPC
Wahpeton 3 steps of 25 MVar Capacitor Banks	\$3,000,000	\$3,946	OTP LPC
Alexandria 115 kV 16.2 MVar Capacitor Bank (OTP Owned)	\$1,000,000	\$808	OTP LPC
Total Cost Per Project:		\$1,662,559	

J840			
Network Upgrade	NU Cost Estimate (\$)	J840 Cost Estimate (\$)	NU Type
Twin Brooks-Alexandria	\$242,400,000	\$994,531	ERIS Voltage
Alexandria 4 steps of 50 MVar Capacitor banks and 345 kV bus expansion	\$26,000,000	\$165,768	ERIS Voltage
Silver Lake 60 MVar Capacitor Bank	\$2,500,000	\$4,604	GRE LPC
Wahpeton 3 steps of 25 MVar Capacitor Banks	\$3,000,000	\$7,191	OTP LPC
Bagley 20 MVar Capacitor Bank	\$1,000,000	\$456	OTP LPC
Alexandria 115 kV 16.2 MVar Capacitor Bank (OTP Owned)	\$1,000,000	\$2,050	OTP LPC
Panther 60 MVar Capacitor Bank	\$2,500,000	\$2,752	GRE LPC
GRE Prairie Voltage Collapse Network Upgrade	\$211,350,000	\$24,827	GRE LPC
CCS GRE LPC Voltage Network Upgrade	\$26,500,000	\$9,104	GRE LPC
Total Cost Per Project:		\$1,211,283	

J869			
Network Upgrade	NU Cost Estimate (\$)	J869 Cost Estimate (\$)	NU Type
Coon Creek Substation Reconfiguration	\$15,000,000	\$15,000,000	ERIS Thermal
Total Cost Per Project:		\$15,000,000	

J873			
Network Upgrade	NU Cost Estimate (\$)	J873 Cost Estimate (\$)	NU Type
Morgan – Tiffin 345 kV Structure and Terminal Equipment Remediation	\$1,500,000	\$1,500,000	ERIS Thermal
Hills – Tiff 345 kV Structure Replacements	\$400,000	\$400,000	ERIS Thermal
Twin Brooks-Alexandria	\$242,400,000	\$497,265	ERIS Voltage
Alexandria 4 steps of 50 MVar Capacitor banks and 345 kV bus expansion	\$26,000,000	\$38,254	ERIS Voltage



Total Cost Per Project:		\$2,435,519	
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J874			
Network Upgrade	NU Cost Estimate (\$)	J874 Cost Estimate (\$)	NU Type
Twin Brooks-Alexandria	\$242,400,000	\$1,989,062	ERIS Voltage
Alexandria 4 steps of 50 MVar Capacitor banks and 345 kV bus expansion	\$26,000,000	\$420,795	ERIS Voltage
Additional Bison Breaker	\$1,500,000	\$3,338	OTP LPC
Silver Lake 60 MVar Capacitor Bank	\$2,500,000	\$28,439	GRE LPC
Wahpeton 3 steps of 25 MVar Capacitor Banks	\$3,000,000	\$22,182	OTP LPC
Audubon 40 MVar Capacitor Bank	\$1,500,000	\$4,518	OTP LPC
Bagley 20 MVar Capacitor Bank	\$1,000,000	\$2,620	OTP LPC
Alexandria 115 kV 16.2 MVar Capacitor Bank (OTP Owned)	\$1,000,000	\$7,547	OTP LPC
Panther 60 MVar Capacitor Bank	\$2,500,000	\$64,495	GRE LPC
GRE Prairie Voltage Collapse Network Upgrade	\$211,350,000	\$92,643	GRE LPC
CCS GRE LPC Voltage Network Upgrade	\$26,500,000	\$9,372	GRE LPC
Total Cost Per Project:		\$2,645,011	

J877			
Network Upgrade	NU Cost Estimate (\$)	J877 Cost Estimate (\$)	NU Type
Twin Brooks-Alexandria	\$242,400,000	\$1,989,062	ERIS Voltage
Alexandria 4 steps of 50 MVar Capacitor banks and 345 kV bus expansion	\$26,000,000	\$293,281	ERIS Voltage
Silver Lake 60 MVar Capacitor Bank	\$2,500,000	\$5,680	GRE LPC
Wahpeton 3 steps of 25 MVar Capacitor Banks	\$3,000,000	\$7,860	OTP LPC
Bagley 20 MVar Capacitor Bank	\$1,000,000	\$683	OTP LPC
Alexandria 115 kV 16.2 MVar Capacitor Bank (OTP Owned)	\$1,000,000	\$2,135	OTP LPC
Panther 60 MVar Capacitor Bank	\$2,500,000	\$3,814	GRE LPC
GRE Prairie Voltage Collapse Network Upgrade	\$211,350,000	\$17,640	GRE LPC
Total Cost Per Project:		\$2,320,155	

J880			
Network Upgrade	NU Cost Estimate (\$)	J880 Cost Estimate (\$)	NU Type
Magic City – Souris Rebuild	\$7,000,000	\$7,000,000	ERIS Thermal
McHenry Transformer Upgrade	\$3,000,000	\$3,000,000	ERIS Thermal



Twin Brooks-Alexandria	\$242,400,000	\$10,157,928	ERIS Voltage
J628 POI-Prairie 230 kV Rebuild	\$16,770,000	\$301,257	ERIS Thermal
J628 POI-J897 POI 230 kV Rebuild	\$28,545,000	\$1,221,761	ERIS Thermal
Alexandria 4 steps of 50 MVar Capacitor banks and 345 kV bus expansion	\$26,000,000	\$1,313,389	ERIS Voltage
Additional Bison Breaker	\$1,500,000	\$97,582	OTP LPC
Silver Lake 60 MVar Capacitor Bank	\$2,500,000	\$125,673	GRE LPC
Jamestown 75 MVar Capacitor Bank	\$4,000,000	\$400,081	OTP LPC
Wahpeton 3 steps of 25 MVar Capacitor Banks	\$3,000,000	\$161,690	OTP LPC
Audubon 40 MVar Capacitor Bank	\$1,500,000	\$91,672	OTP LPC
Perham 2 steps of 20 MVar Capacitor Banks	\$2,000,000	\$120,607	OTP LPC
Bagley 20 MVar Capacitor Bank	\$1,000,000	\$56,901	OTP LPC
Alexandria 115 kV 16.2 MVar Capacitor Bank (OTP Owned)	\$1,000,000	\$44,490	OTP LPC
Panther 60 MVar Capacitor Bank	\$2,500,000	\$105,942	GRE LPC
GRE Prairie Voltage Collapse Network Upgrade	\$211,350,000	\$9,793,972	GRE LPC
CCS GRE LPC Voltage Network Upgrade	\$26,500,000	\$2,863,137	GRE LPC
Total Cost Per Project:		\$36,856,082	

J885			
Network Upgrade	NU Cost Estimate (\$)	J801 Cost Estimate (\$)	NU Type
	\$0	\$0	
Total Cost Per Project:		\$0	

J889			
Network Upgrade	NU Cost Estimate (\$)	J889 Cost Estimate (\$)	NU Type
Devil's Lake-Ramsey Rebuild	\$450,000	\$199,056	ERIS Thermal
Twin Brooks-Alexandria	\$242,400,000	\$11,070,226	ERIS Voltage
J628 POI-Prairie 230 kV Rebuild	\$16,770,000	\$2,699,267	ERIS Thermal
J628 POI-J897 POI 230 kV Rebuild	\$28,545,000	\$10,946,982	ERIS Thermal
J889 POI-Ramsey 230 kV Rebuild	\$46,290,000	\$20,422,059	ERIS Thermal
Alexandria 4 steps of 50 MVar Capacitor banks and 345 kV bus expansion	\$26,000,000	\$1,185,875	ERIS Voltage
New J628-Prairie 230 kV 2 nd Circuit	\$22,360,000	\$6,440,380	ERIS Voltage



Wilton-Bemidji Uprate	\$50,000	\$9,222	ERIS Thermal
Additional Bison Breaker	\$1,500,000	\$90,847	OTP LPC
Silver Lake 60 MVar Capacitor Bank	\$2,500,000	\$90,688	GRE LPC
Jamestown 75 MVar Capacitor Bank	\$4,000,000	\$191,881	OTP LPC
Wahpeton 3 steps of 25 MVar Capacitor Banks	\$3,000,000	\$108,724	OTP LPC
Audubon 40 MVar Capacitor Bank	\$1,500,000	\$72,842	OTP LPC
Perham 2 steps of 20 MVar Capacitor Banks	\$2,000,000	\$187,261	OTP LPC
Bagley 20 MVar Capacitor Bank	\$1,000,000	\$92,664	OTP LPC
Alexandria 115 kV 16.2 MVar Capacitor Bank (OTP Owned)	\$1,000,000	\$34,055	OTP LPC
Panther 60 MVar Capacitor Bank	\$2,500,000	\$80,918	GRE LPC
Ramsey Transformer Upgrade	\$5,000,000	\$5,000,000	GRE LPC
GRE Prairie Voltage Collapse Network Upgrade	\$211,350,000	\$26,960,145	GRE LPC
Total Cost Per Project:		\$85,883,092	

J897			
Network Upgrade	NU Cost Estimate (\$)	J897 Cost Estimate (\$)	NU Type
Devil's Lake-Ramsey Rebuild	\$450,000	\$250,944	ERIS Thermal
Twin Brooks-Alexandria	\$242,400,000	\$13,556,553	ERIS Voltage
J628 POI-Prairie 230 kV Rebuild	\$16,770,000	\$4,037,424	ERIS Thermal
J628 POI-J897 POI 230 kV Rebuild	\$28,545,000	\$16,376,257	ERIS Thermal
J889 POI-Ramsey 230 kV Rebuild	\$46,290,000	\$25,867,941	ERIS Thermal
Alexandria 4 steps of 50 MVar Capacitor banks and 345 kV bus expansion	\$26,000,000	\$930,848	ERIS Voltage
New J628-Prairie 230 kV 2 nd Circuit	\$22,360,000	\$7,265,749	ERIS Voltage
Wilton-Bemidji Uprate	\$50,000	\$12,568	ERIS Thermal
Wilton-Bemidji 115 kV Uprate (OTP LPC)	\$50,000	\$15,385	OTP LPC
Additional Bison Breaker	\$1,500,000	\$112,911	OTP LPC
Silver Lake 60 MVar Capacitor Bank	\$2,500,000	\$106,981	GRE LPC
Jamestown 75 MVar Capacitor Bank	\$4,000,000	\$201,866	OTP LPC
Wahpeton 3 steps of 25 MVar Capacitor Banks	\$3,000,000	\$125,175	OTP LPC
Audubon 40 MVar Capacitor Bank	\$1,500,000	\$87,430	OTP LPC
Perham 2 steps of 20 MVar Capacitor Banks	\$2,000,000	\$253,127	OTP LPC
Bagley 20 MVar Capacitor Bank	\$1,000,000	\$126,203	OTP LPC
Alexandria 115 kV 16.2 MVar Capacitor Bank (OTP Owned)	\$1,000,000	\$41,012	OTP LPC
Panther 60 MVar Capacitor Bank	\$2,500,000	\$97,343	GRE LPC



GRE Prairie Voltage Collapse Network Upgrade	\$211,350,000	\$39,650,435	GRE LPC
Total Cost Per Project:		\$109,116,152	

J898			
Network Upgrade	NU Cost Estimate (\$)	J898 Cost Estimate (\$)	NU Type
Beaver Creek-Harmony 161 kV Uprate	\$2,000,000	\$2,000,000	DPC LPC
Wabaco-Alma 161 kV Uprate	\$200,000	\$200,000	DPC LPC
Total Cost Per Project:		\$2,200,000	

J901			
Network Upgrade	NU Cost Estimate (\$)	J901 Cost Estimate (\$)	NU Type
Twin Brooks-Alexandria	\$242,400,000	\$2,486,327	ERIS Voltage
Alexandria 4 steps of 50 MVar Capacitor banks and 345 kV bus expansion	\$26,000,000	\$1,440,902	ERIS Voltage
Additional Bison Breaker	\$1,500,000	\$2,101	OTP LPC
Silver Lake 60 MVar Capacitor Bank	\$2,500,000	\$128,742	GRE LPC
Wahpeton 3 steps of 25 MVar Capacitor Banks	\$3,000,000	\$30,461	OTP LPC
Audubon 40 MVar Capacitor Bank	\$1,500,000	\$5,404	OTP LPC
Bagley 20 MVar Capacitor Bank	\$1,000,000	\$3,827	OTP LPC
Alexandria 115 kV 16.2 MVar Capacitor Bank (OTP Owned)	\$1,000,000	\$27,145	OTP LPC
Panther 60 MVar Capacitor Bank	\$2,500,000	\$392,078	GRE LPC
Total Cost Per Project:		\$4,516,987	

J905			
Network Upgrade	NU Cost Estimate (\$)	J1155 Cost Estimate (\$)	NU Type
Twin Brooks-Alexandria	\$242,400,000	\$994,531	ERIS Voltage
Alexandria 4 steps of 50 MVar Capacitor banks and 345 kV bus expansion	\$26,000,000	\$165,768	ERIS Voltage
Additional Bison Breaker	\$1,500,000	\$1,590	OTP LPC
Silver Lake 60 MVar Capacitor Bank	\$2,500,000	\$12,026	GRE LPC
Wahpeton 3 steps of 25 MVar Capacitor Banks	\$3,000,000	\$9,666	OTP LPC
Audubon 40 MVar Capacitor Bank	\$1,500,000	\$1,999	OTP LPC
Bagley 20 MVar Capacitor Bank	\$1,000,000	\$1,081	OTP LPC
Alexandria 115 kV 16.2 MVar Capacitor Bank (OTP Owned)	\$1,000,000	\$3,367	OTP LPC
Panther 60 MVar Capacitor Bank	\$2,500,000	\$26,588	GRE LPC



GRE Prairie Voltage Collapse Network Upgrade	\$211,350,000	\$38,538	GRE LPC
CCS GRE LPC Voltage Network Upgrade	\$26,500,000	\$3,703	GRE LPC
Total Cost Per Project:		\$1,258,857	

J916			
Network Upgrade	NU Cost Estimate (\$)	J801 Cost Estimate (\$)	NU Type
	\$0	\$0	
Total Cost Per Project:		\$0	

J926			
Network Upgrade	NU Cost Estimate (\$)	J801 Cost Estimate (\$)	NU Type
	\$0	\$0	
Total Cost Per Project:		\$0	

J927			
Network Upgrade	NU Cost Estimate (\$)	J927 Cost Estimate (\$)	NU Type
Twin Brooks-Alexandria	\$242,400,000	\$497,265	ERIS Voltage
Walters-Alden 69 kV Rebuild	\$4,700,000	\$4,700,000	ERIS Thermal
Alden-Aleawst 69 kV Rebuild	\$5,100,00	\$5,100,000	ERIS Thermal
Total Cost Per Project:		\$10,297,265	

J929			
Network Upgrade	NU Cost Estimate (\$)	J929 Cost Estimate (\$)	NU Type
New Mandon-J580 230 kV Line	\$28,000,000	\$455,515	ERIS Thermal/ERIS Voltage
Rebuild Merricourt – Wishek 230 kV	\$5,800,000	\$469,581.31	ERIS Thermal
Merricourt – Tatanka Structure Remediation	\$100,000	\$8,330.12	ERIS Thermal
Twin Brooks-Alexandria	\$242,400,000	\$1,124,917.96	ERIS Voltage
Alexandria 4 steps of 50 MVar Capacitor banks and 345 kV bus expansion	\$26,000,000	\$165,767.53	ERIS Voltage
Additional Merricourt Breakers	\$1,500,000	\$266,110	MDU LPC
Foxtail-Tatanka North 230 kV Upgrade	\$2,000,000	\$336,596	MDU LPC
Foxtail-Ellendale 230 kV Rebuild	\$10,500,000	\$1,767,131	MDU LPC
Merricourt-Tatanka North 230 kV Upgrade	\$1,000,000	\$168,298	MDU LPC
Merricourt-Ellendale 230 kV Reconnector	\$18,000,000	\$3,029,440	MDU LPC
Additional Bison Breaker	\$1,500,000	\$5,980	OTP LPC
Silver Lake 60 MVar Capacitor Bank	\$2,500,000	\$19,268	GRE LPC
Jamestown 75 MVar Capacitor Bank	\$4,000,000	\$14,979	OTP LPC
Wahpeton 3 steps of 25 MVar Capacitor Banks	\$3,000,000	\$28,423	OTP LPC
Audubon 40 MVar Capacitor Bank	\$1,500,000	\$5,908	OTP LPC
Perham 2 steps of 20 MVar Capacitor Banks	\$2,000,000	\$3,186	OTP LPC
Bagley 20 MVar Capacitor Bank	\$1,000,000	\$2,976	OTP LPC
Alexandria 115 kV 16.2 MVar Capacitor Bank (OTP Owned)	\$1,000,000	\$5,212	OTP LPC
Panther 60 MVar Capacitor Bank	\$2,500,000	\$11,148	GRE LPC
GRE Prairie Voltage Collapse Network Upgrade	\$211,350,000	\$258,676	GRE LPC
CCS GRE LPC Voltage Network Upgrade	\$26,500,000	\$135,395	GRE LPC
Total Cost Per Project:		\$8,282,838	

J933			
Network Upgrade	NU Cost Estimate (\$)	J933 Cost Estimate (\$)	NU Type
Twin Brooks-Alexandria	\$242,400,000	\$53,706,421	ERIS Voltage
Alexandria 4 steps of 50 MVar Capacitor banks and 345 kV bus expansion	\$26,000,000	\$2,333,497	ERIS Voltage
Ellendale-Aberdeen Jct 115 kV Uprate	\$60,000	\$60,000	MDU LPC
Canby-Granite Falls 115 kV Rebuild	\$17,700,000	\$8,045,455	OTP LPC
Wahpeton-Fergus Falls 230 kV Uprate	\$1,150,000	\$566,925	OTP LPC



Additional Bison Breaker	\$1,500,000	\$20,567	OTP LPC
Silver Lake 60 MVar Capacitor Bank	\$2,500,000	\$204,699	GRE LPC
Wahpeton 3 steps of 25 MVar Capacitor Banks	\$3,000,000	\$244,093	OTP LPC
Audubon 40 MVar Capacitor Bank	\$1,500,000	\$41,988	OTP LPC
Bagley 20 MVar Capacitor Bank	\$1,000,000	\$23,204	OTP LPC
Alexandria 115 kV 16.2 MVar Capacitor Bank (OTP Owned)	\$1,000,000	\$140,102	OTP LPC
Panther 60 MVar Capacitor Bank	\$2,500,000	\$215,463	GRE LPC
GRE Prairie Voltage Collapse Network Upgrade	\$211,350,000	\$593,400	GRE LPC
Total Cost Per Project:		\$66,195,814	

J946			
Network Upgrade	NU Cost Estimate (\$)	J946 Cost Estimate (\$)	NU Type
Sheyenne-Lake Park Uprate	\$1,300,000	\$95,991	ERIS Thermal
Sheyenne-Mapleton Rebuild	\$17,600,000	\$3,510,105	ERIS Thermal
Buffalo Transformer #1 Upgrade	\$3,000,000	\$1,002,468	ERIS Thermal
Buffalo Transformer #2 Upgrade	\$3,000,000	\$1,002,468	ERIS Thermal
Mapleton-CSLTNET Uprate	\$25,000	\$7,218	ERIS Thermal
Badura-Hubbard Uprate	\$600,000	\$140,229	ERIS Thermal
Twin Brooks-Alexandria	\$242,400,000	\$7,458,981	ERIS Voltage
Alexandria 4 steps of 50 MVar Capacitor banks and 345 kV bus expansion	\$26,000,000	\$1,058,362	ERIS Voltage
Wahpeton-Fergus Falls 230 kV Uprate	\$1,150,000	\$194,358	OTP LPC
Audubon-Hubbard 230 kV Uprate (OTP LPC)	\$250,000	\$83,333	OTP LPC
Additional Bison Breaker	\$1,500,000	\$103,694	OTP LPC
Silver Lake 60 MVar Capacitor Bank	\$2,500,000	\$116,825	GRE LPC
Wahpeton 3 steps of 25 MVar Capacitor Banks	\$3,000,000	\$156,914	OTP LPC
Audubon 40 MVar Capacitor Bank	\$1,500,000	\$128,196	OTP LPC
Perham 2 steps of 20 MVar Capacitor Banks	\$2,000,000	\$90,465	OTP LPC
Bagley 20 MVar Capacitor Bank	\$1,000,000	\$31,009	OTP LPC
Alexandria 115 kV 16.2 MVar Capacitor Bank (OTP Owned)	\$1,000,000	\$65,315	OTP LPC
Panther 60 MVar Capacitor Bank	\$2,500,000	\$61,863	GRE LPC
GRE Prairie Voltage Collapse Network Upgrade	\$211,350,000	\$149,134	GRE LPC
Total Cost Per Project:		\$15,456,928	

1.3. Total Network Upgrades

The cost allocation of Network Upgrades for the projects in the DPP 2017 August West Phase 1 reflects responsibilities for mitigating system impacts. The total cost of network upgrades is listed in Table 2 below. The costs for Network Upgrades are planning-level estimates and subject to revision in the facility studies.

Table 2: Total Cost of Network Upgrades for DPP 2017 August West Phase 1 Projects

Project	ERIS Network Upgrades (\$)					NRIS Network Upgrades (\$)	Interconnection Facilities (\$)		Shared Network Upgrades (\$)	Total Network Upgrade Cost for Milestone Calculation (\$)	M2 (\$)	M3 (\$)
	Steady State	LPC Studies	Stability	Short Circuit	Affected System	Deliverability	TO Network Upgrades	TO – Owned Direct Assigned				
a	b	c	d	e	f	g	h	i	j	k = b+c+d+e+g+h+i	\$10,000/MW	10% of (j) from Phase I- M2
G517	\$45,472,105	\$147,626	Phase 2	Phase 2	Phase 2	\$0	\$263,000		\$0	\$45,882,731	\$562,821	\$4,025,452
J447	\$522,768	\$0	Phase 2	Phase 2	Phase 2	\$0	\$944,000		\$0	\$1,466,768	\$600,000	\$0
J545	\$4,284,193	\$542,065	Phase 2	Phase 2	Phase 2	\$0	\$1,000,000		\$0	\$5,826,258	\$600,000	\$0
J580	\$60,355,021	\$45,912,848	Phase 2	Phase 2	Phase 2	\$0	\$6,500,000		\$2,297,258	\$115,065,127	\$1,223,600	\$10,282,913
J628	\$49,247,093	\$97,528,330	Phase 2	Phase 2	Phase 2	\$0	\$5,500,000		\$0	\$152,275,423	\$1,600,000	\$13,627,542
J705	\$10,124,396	\$10,114,644	Phase 2	Phase 2	Phase 2	\$0	\$1,643,444		\$0	\$21,882,484	\$400,000	\$1,788,248
J706	\$10,124,396	\$10,114,644	Phase 2	Phase 2	Phase 2	\$0	\$1,781,300		\$0	\$22,020,340	\$400,000	\$1,802,034
J713	\$26,753,627	\$30,343,932	Phase 2	Phase 2	Phase 2	\$0	\$1,188,556		\$0	\$58,286,115	\$1,200,000	\$4,628,612
J720	\$2,282,343	\$69,436	Phase 2	Phase 2	Phase 2	\$0	\$9,270,000		\$0	\$11,621,779	\$800,000	\$362,178
J722	\$14,012,839	\$11,187,004	Phase 2	Phase 2	Phase 2	\$0	\$1,250,000		\$194,284	\$26,644,127	\$800,000	\$1,864,413
J743	\$44,639,376	\$2,362,214	Phase 2	Phase 2	Phase 2	\$0	\$5,000,000		\$0	\$52,001,590	\$800,000	\$4,400,159
J785	\$1,160,298	\$15,834	Phase 2	Phase 2	Phase 2	\$0	\$9,270,000		\$0	\$10,446,132	\$420,000	\$624,613
J801	\$0	\$0	Phase 2	Phase 2	Phase 2	\$0	\$3,400,000		\$0	\$3,400,000	\$296,000	\$44,000
J803	\$663,033	\$50,338	Phase 2	Phase 2	Phase 2	\$0	\$500,000		\$0	\$1,213,371	\$130,000	\$0
J816	\$6,194,602	\$418,495	Phase 2	Phase 2	Phase 2	\$0	\$1,250,000		\$0	\$7,863,097	\$240,000	\$546,310
J836	\$1,657,564	\$4,995	Phase 2	Phase 2	Phase 2	\$0	\$1,900,000		\$0	\$3,562,559	\$800,000	\$0
J840	\$1,160,298	\$50,984	Phase 2	Phase 2	Phase 2	\$0	\$13,350,000		\$0	\$14,561,282	\$600,000	\$856,128
J869	\$15,000,000	\$0	Phase 2	Phase 2	Phase 2	\$0	\$3,500,000		\$0	\$18,500,000	\$200,000	\$1,650,000
J873	\$2,435,519	\$0	Phase 2	Phase 2	Phase 2	\$0	\$13,350,000		\$0	\$15,785,519	\$800,000	\$778,552
J874	\$2,409,856	\$235,154	Phase 2	Phase 2	Phase 2	\$0	\$8,000,000		\$0	\$10,645,010	\$600,000	\$464,501
J877	\$2,282,343	\$37,812	Phase 2	Phase 2	Phase 2	\$0	\$50,000		\$0	\$2,370,155	\$1,000,000	\$0
J880	\$22,994,336	\$13,861,747	Phase 2	Phase 2	Phase 2	\$0	\$8,000,000		\$0	\$44,856,083	\$600,000	\$3,885,608
J885	\$0	\$0	Phase 2	Phase 2	Phase 2	\$0	\$944,000		\$0	\$944,000	\$256,000	\$0

Project	ERIS Network Upgrades (\$)					NRIS Network Upgrades (\$)	Interconnection Facilities (\$)		Shared Network Upgrades (\$)	Total Network Upgrade Cost for Milestone Calculation (\$)	M2 (\$)	M3 (\$)
	Steady State	LPC Studies	Stability	Short Circuit	Affected System	Deliverability	TO Network Upgrades	TO – Owned Direct Assigned				
a	b	c	d	e	f	g	h	i	j	k = b+c+d+e+g+h+i	\$10,000/MW	10% of (j) from Phase I-M2
J889	\$52,973,067	\$32,910,025	Phase 2	Phase 2	Phase 2	\$0	\$5,500,000		\$0	\$91,383,092	\$600,000	\$8,538,309
J897	\$68,298,285	\$40,817,868	Phase 2	Phase 2	Phase 2	\$0	\$5,500,000		\$0	\$114,616,153	\$760,000	\$10,701,615
J898	\$0	\$2,200,000	Phase 2	Phase 2	Phase 2	\$0	\$4,300,000		\$0	\$6,500,000	\$400,000	\$250,000
J901	\$3,927,229	\$589,758	Phase 2	Phase 2	Phase 2	\$0	\$13,000,000		\$0	\$17,516,987	\$800,000	\$951,699
J905	\$1,160,298	\$98,558	Phase 2	Phase 2	Phase 2	\$0	\$1,250,000		\$0	\$2,508,856	\$160,000	\$90,886
J916	\$0	\$0	Phase 2	Phase 2	Phase 2	\$0	N/A		\$0	\$0	\$8,000	\$0
J926	\$0	\$0	Phase 2	Phase 2	Phase 2	\$0	\$11,000,000		\$0	\$11,000,000	\$405,120	\$694,880
J927	\$10,297,265	\$0	Phase 2	Phase 2	Phase 2	\$0	\$263,000		\$0	\$10,560,265	\$400,000	\$656,027
J929	\$2,224,112	\$6,058,726	Phase 2	Phase 2	Phase 2	\$0	\$550,000		\$509,970	\$9,342,808	\$100,000	\$834,281
J933	\$56,039,917	\$10,155,896	Phase 2	Phase 2	Phase 2	\$0	\$11,000,000		\$0	\$77,195,813	\$800,000	\$6,919,581
J946	\$14,275,821	\$1,181,106	Phase 2	Phase 2	Phase 2	\$0	\$5,000,000		\$0	\$20,456,927	\$800,000	\$1,245,693
Total	\$532,972,000	\$317,010,039				\$0	\$155,217,300		\$3,001,512	\$1,008,200,851	\$20,161,541	\$82,514,234

The preliminary System Impact Study (SIS) for the DPP 2017 August West was performed using models that were developed from the MTEP 18 series models including all prior-queued projects and their associated network upgrades known through the DPP February West Phase 2 analysis. Analyses performed demonstrate the following transmission facilities are required to reliably interconnect this group of generators to the transmission system. Energy Resource Interconnection Service (ERIS) Network Upgrades and Network Resource Interconnection Service (NRIS) Network Upgrades are shown in Table 3. Shared Network Upgrades are shown in Table 4.

Table 3: ERIS & NRIS Upgrades (Planning level cost estimates)

Network Upgrade	TO	GI projects requiring upgrade for ERIS	GI projects requiring upgrade for NRIS	Cost of solution (\$)
New Mandan – J580 230 kV Line	MDU	J580, J929		\$28,000,000
Merricourt-Wishek 230 kV Rebuild	MDU	J580, J929		\$5,800,000
Merricourt-Tatanka 230 kV Uprate	MDU	J580, J929		\$100,000
Coon Creek Substation Reconfiguration	XEL	J869		\$15,000,000
Sheyenne-Lake Park 230 kV Uprate	XEL, OTP, MPC	J580, J713, J743, J816, J946		\$1,300,000
Sheyenne-Mapleton 115 kV Rebuild	XEL, OTP	J743, J816, J946		\$17,600,000

Network Upgrade	TO	GI projects requiring upgrade for ERIS	GI projects requiring upgrade for NRIS	Cost of solution (\$)
Magic City-Souris 115 kV Rebuild	XEL	J880		\$7,000,000
Granite City-Crossroads 115 kV Rebuild	XEL	J743		\$3,700,000
Crossroads-MEI International 115 kV Rebuild	XEL	J743		\$1,300,000
MEI International-Westwood 115kV Rebuild	XEL	J743		\$2,000,000
West St. Cloud-Westwood 115 kV Rebuild	XEL	J743		\$600,000
Buffalo 345/115 kV Transformer #1 Upgrade	OTP	J743, J946		\$3,000,000
Buffalo 345/115 kV Transformer #2 Upgrade	OTP	J743, J946		\$3,000,000
Mapleton-CSLTNET 115 kV Uprate	OTP	J743, J816, J946		\$25,000
Mahnomn-Ulrich 115 kV Uprate	OTP	J628		\$50,000
Winger-Fertile 115 kV Uprate	OTP	J628		\$100,000
Wilton-Bemidji 115 kV Uprate	OTP	J628, J889, J897		\$50,000
Crookston-Fertile 115 kV Uprate	OTP	J628		\$250,000
Crookston-Falconer 115 kV Uprate	OTP	J628		\$25,000
Devil's Lake-Ramsey 115 kV Rebuild	OTP	J889, J897		\$450,000
Heron Lake 161/69 kV Transformer #1 Upgrade	ITCM	G517		\$2,520,000
Heron Lake 161/69 kV Transformer #2 Upgrade	ITCM	G517		\$2,520,000
Morgan Vallley-Tiffin 345 kV Uprate	MEC	J873		\$1,500,000
Hills-Tiffin 345 kV Uprate	MEC	J873		\$400,000
Badura-Hubbard 230 kV Uprate	MP, GRE	J743, J816, J946		\$600,000
Riverton-Mud Lake 230 kV Uprate	GRE	J628		\$882,000
McHenry 230/115 kV Transformer #1 Upgrade	GRE	J880		\$3,000,000
Hubbard-Audubon 230 kV Uprate	MP, GRE, XEL, OTP	J743		\$2,000,000
New Twin Brooks-Alexandria 345 kV Line	OTP, MRES	G517, J447, J545, J580, J628, J705, J706, J713, J720, J722, J743, J785, J803, J816, J836, J840, J873, J874, J877, J880, J889, J897, J901, J905, J927, J929, J933, J946		\$242,400,000
Wanda-Sundown 69 kV Rebuild	GRE	G517		\$3,150,000
Wanda-Waterbury 69 kV Rebuild	GRE	G517		\$1,500,000
Windms-Wilder 69 kV Uprate	GRE	G517		\$100,000

Network Upgrade	TO	GI projects requiring upgrade for ERI	GI projects requiring upgrade for NRIS	Cost of solution (\$)
Walters-Alden 69 kV Rebuild	ITCM	J927		\$4,700,000
Alden-Aleawst 69 kV Rebuild	ITCM	J927		\$5,100,000
Cottonwood-Waterbury 69 kV Rebuild	ITCM	G517		\$4,400,000
Cottonwood-Highway Tap 69 kV Rebuild	ITCM	G517		\$725,000
Highway Tap-Lambrtn 69 kV Rebuild	ITCM	G517		\$334,000
Jefferson Tap-N Storden 69 kV Rebuild	ITCM	G517		\$2,900,000
Jefferson Tap-Storden 69 kV Rebuild	ITCM	G517		\$900,000
N Storden-Lambrtn 69 kV Rebuild	ITCM	G517		\$5,900,000
S Storden-Heron Lake 69 kV Rebuild	ITCM	G517		\$8,400,000
S Storden-Storden 69 kV Rebuild	ITCM	G517		\$4,000,000
Springfield-Dotson Corners 69 kV Uprate	ITCM/GRE	G517		\$60,000
Brookville-Sundown 69 kV Rebuild	GRE	G517		\$2,700,000
Brookville-Cobden Tap 69 kV Rebuild	GRE/ITCM	G517		\$2,570,000
Springfield-Cobden Tap 69 kV Rebuild	ITCM	G517		\$396,000
New J628 POI- Prairie 230 kV 2 nd Circuit	GRE	J628, J889, J897		\$22,360,000
J628 POI-Prairie 230 kV Rebuild	GRE	J628, J880, J889, J897		\$16,770,000
J628 POI- J897 POI 230 kV Rebuild	GRE	J880, J889, J897		\$28,545,000
J889 POI-Ramsey 230 kV Rebuild	GRE	J889, J897		\$46,290,000
Alexandria four 50 MVar Capacitor Banks and 345 kV bus expansion	MRES	G517, J447, J545, J580, J628, J705, J706, J713, J720, J722, J743, J785, J803, J816, J836, J840, J873, J874, J877, J880, J889, J897, J901, J905, J929, J933, J946		\$26,000,000
Beaver Creek-Harmony 161 kV Uprate	DPC	J898		\$2,000,000
Wabaco-Alma 161 kV Uprate	DPC	J898		\$200,000
Ellendale-Aberdean Jct 115 kV Uprate	MDU	J933		\$60,000
Fox Tail-Tatanka North 230 kV Upgrade	MDU	J580, J929		\$2,000,000
Fox Tail-Ellendale 230 kV Rebuild	MDU	J580, J929		\$10,500,000
Merricourt-Tatanka North 230 KV Reconnector	MDU	J580, J929		\$1,000,000
Merricourt-Ellendale 230 kV Rebuild	MDU	J580, J929		\$18,000,000
Additional Merricourt Breakers	MDU	J580, J929		\$1,500,000

Network Upgrade	TO	GI projects requiring upgrade for ERI	GI projects requiring upgrade for NRIS	Cost of solution (\$)
Canby-Granite Falls 115 kV Rebuild	OTP	J722, J933		\$17,700,000
Winger-Fertile 115 kV Uprate (OTP LPC)	OTP	J628		\$450,000
Wilton-Bemidji 115 kV Uprate (OTP LPC)	OTP	J628, J897		\$50,000
Crookston-Fertile 115 kV Uprate (OTP LPC)	OTP	J628		\$250,000
Donalds-Warsaw 115 kV Uprate	OTP	J628		\$50,000
Wahpeton-Fergus Falls 230 kV Uprate	OTP	J743, J933, J946		\$1,150,000
Audubon-Hubbard 230 kV Uprate (OTP LPC)	OTP	J743, J946		\$250,000
Additional Bison Breaker	XEL	G517, J545, J580, J628, J705, J706, J713, J722, J743, J803, J816, J874, J880, J889, J897, J901, J905, J929, J933, J946		\$1,500,000
Silver Lake 60 MVar Capacitor Bank	GRE	G517, J545, J580, J628, J705, J706, J713, J720, J722, J743, J785, J803, J816, J836, J840, J874, J877, J880, J889, J897, J901, J905, J929, J933, J946		\$2,500,000
Jamestown 75 MVar Capacitor Bank	OTP	J580, J628, J705, J706, J713, J880, J889, J897, J929		\$4,000,000
Wahpeton 3 steps of 25 MVar Capacitor Banks	OTP	G517, J545, J580, J628, J705, J706, J713, J720, J722, J743, J785, J803, J816, J836, J840, J874, J877, J880, J889, J897, J901, J905, J929, J933, J946		\$3,000,000
Audubon 40 MVar Capacitor Bank	OTP	G517, J545, J580, J628, J705, J706,		\$1,500,000

Network Upgrade	TO	GI projects requiring upgrade for ERIIS	GI projects requiring upgrade for NRIS	Cost of solution (\$)
		J713, J720, J722, J743, J803, J816, J874, J880, J889, J897, J901, J905, J929, J933, J946		
Perham 2 steps of 20 MVar Capacitor Banks	OTP	J580, J628, J705, J706, J713, J743, J816, J880, J889, J897, J929, J946		\$2,000,000
Bagley 20 MVar Capacitor Bank	OTP	G517, J545, J580, J628, J705, J706, J713, J720, J722, J743, J785, J803, J816, J840, J874, J877, J880, J889, J897, J901, J905, J929, J933, J946		\$1,000,000
Alexandria 115 kV 16.2 MVar Capacitor Bank (OTP Owned)	OTP	G517, J545, J580, J628, J705, J706, J713, J720, J722, J743, J785, J803, J816, J836, J840, J874, J877, J880, J889, J897, J901, J905, J929, J933, J946		\$1,000,000
Panther 60 MVar Capacitor Bank	GRE	G517, J545, J580, J628, J705, J706, J713, J720, J722, J743, J785, J803, J816, J840, J874, J877, J880, J889, J897, J901, J905, J929, J933, J946		\$2,500,000
Ramsey Transformer Upgrade	GRE	J889		\$5,000,000
J628-Bison 345 kV Line*	GRE	G517, J545, J580, J628, J705, J706,		\$198,000,000

Network Upgrade	TO	GI projects requiring upgrade for ERI	GI projects requiring upgrade for NRIS	Cost of solution (\$)
		J713, J720, J722, J743, J803, J816, J840, J874, J877, J880, J889, J897, J905, J929, J933, J946		
J628 345/230 kV Transformer*	GRE	G517, J545, J580, J628, J705, J706, J713, J720, J722, J743, J803, J816, J840, J874, J877, J880, J889, J897, J905, J929, J933, J946		\$9,200,000
J628 345 kV Substation*	GRE	G517, J545, J580, J628, J705, J706, J713, J720, J722, J743, J803, J816, J840, J874, J877, J880, J889, J897, J905, J929, J933, J946		\$4,150,000
Jamestown +/-75 MVar SVC**	OTP	J545, J580, J705, J706, J713, J840, J874, J880, J905, J929		\$20,000,000
River Road two steps of 75 MVar Capacitor Banks**	GRE	J545, J580, J705, J706, J713, J840, J874, J880, J905, J929		\$5,000,000
Additional Audubon 40 MVar Capacitor Bank**	OTP	J545, J580, J705, J706, J713, J840, J874, J880, J905, J929		\$1,500,000

*GRE Prairie Voltage Collapse Network Upgrade

**CCS GRE LPC Voltage Network Upgrade

Table 4: Shared Network Upgrades (Planning level cost estimates)

Shared Network Upgrade	TO	Higher queued projects associated with SNU	Study projects associated with SNU	Cost of solution (\$)
Forman – Oakes 230 kV Uprate (Aug-2015 Cycle)	OTP	J436, J437, J488, G359R	J580, J929	\$300,000
Forman – Oakes 230 kV Uprate (Feb-2016 Cycle)	OTP	J488, G359R	J580, J929	\$950,000
Forman – Oakes 230 kV Uprate (Merricourt Cycle)	OTP	G359R	J580, J929	\$112,750
Forman – Oakes 230 kV Reconductor (Aug-2016 Cycle)	OTP	J302, J503	J580, J929	
Big Stone – Blair 230 kV Uprate (Feb-2016 Cycle)	OTP, NWE	J488, J493, J526	J722	\$150,183
Big Stone – Browns Valley 230 kV Uprate (Feb-2016 Cycle)	OTP	J488, J493, J526	J722	\$285,086

Note:

- 1) Details pertaining to upgrades, costs, and the execution plan for interconnection of the generating facility at the POI will be documented in the Facility Study for Interconnecting Generator.
- 2) Facilities that have been included as base case assumptions and the level of interconnection service that would be conditional upon these facilities being in service will be documented in the GIA (Generator Interconnection Agreement) for each respective GI request successfully achieving GIA execution.
- 3) Analysis performed shows that projects J580, J722, and J929 met the criteria for Shared Network Upgrade cost allocation.

2. FERC Order 827 Compliance Review

The Final Rule of FERC Order 827 “Reactive Power Requirements for Non-Synchronous Generation”, which was issued June 16, 2016, stated that “Under this Final Rule, newly interconnecting non-synchronous generators that have not yet executed a Facilities Study Agreement as of the effective date of this Final Rule will be required to provide dynamic reactive power within the range of 0.95 leading to 0.95 lagging at the high-side of the generator substation.” As such, this Final Rule applies to all non-synchronous projects included in the DPP 2017 August West study cycle.

In this study, the power factor at the high-side of the generator substation for each project was calculated and reviewed. The study method is to set Qgen of each study project at its Qmax, solve the case, then record the P and Q injection on the high side of the generator substation to calculate the lagging power factor (injecting VAR to the system). The same process is then repeated by setting Qgen at Qmin to calculate the leading power factor (absorbing VAR from the system).

The results show that not all projects meet FERC Order 827 requirements. Additional reactive support will be needed for these projects to meet the FERC requirement on reactive power capability prior to the completion of their GIA.

Table 5: FERC Order 827 Review Results

Project	Pmax (MW)	Reactive Power Capability (MVar)	Proposed VAR Compensation	VAR Injection			VAR Absorption			Meet FERC Order 827 Requirement?	Add'l VAR Needed (MVar)
				P (MW)	Q (MVar)	Lagging p.f (pu)	P (MW)	Q (MVar)	Leading p.f (pu)		
G517	130	59.074 -44.442	2 x 9 MVar Cap	128.0	74.4	0.865	124.9	-68.1	-0.878	Yes	
J447	150	30.75 -43.725	3 x 15 MVar Cap	146.6	59.4	0.927	144.9	-85.0	-0.863	Yes	
J545	110	±37	3 x 6 Mvar Cap	108.4	51.3	0.904	107.9	-50.2	-0.907	Yes	
J580	244	112.404 -80.736	N/A	291.4	69.0	0.973	285.5	-228.5	-0.781	No	26.8
J628	400	±133	6 x 8 MVar Cap	397.5	128.4	0.952	396.2	-145.5	-0.939	No	2.3
J705	105	±50.893	N/A	103.1	34.1	0.949	101.9	-81.9	-0.779	Yes	
J706	105	±50.893	N/A	103.0	36.9	0.941	101.9	-76.4	-0.800	Yes	
J713	300	±100	6 x 6 Mvar Cap	297.0	80.6	0.965	295.3	-185.4	-0.847	No	17.0
J720	200	±66	N/A	199.2	27.9	0.990	198.9	-117.8	-0.860	No	37.6
J722	200	±67	3 x 8 MVar Cap	198.6	56.7	0.962	198.0	-114.9	-0.865	No	8.6
J743	210	±91.5	2 x 20 MVar Cap	208.1	100.4	0.901	289.6	-244.1	-0.765	Yes	
J785	105	±34.5120	3 x 10 MVar Cap	104.0	37.8	0.940	103.0	-91.9	-0.746	Yes	
J801	74	±36.4	N/A	73.1	29.1	0.929	72.8	-46.2	-0.844	Yes	
J803	33	±13.25	N/A	32.5	8.6	0.967	32.4	-21.6	-0.832	No	2.1
J816	62	±20	2 x 4 MVar Cap	61.3	18.9	0.956	61.0	-34.3	-0.872	No	1.2
J836	205	±95.5	N/A	201.7	56.0	0.964	200.2	-169.0	-0.764	No	10.3
J840	150	30.75 -43.725	3 x 15 MVar Cap	143.6	48.6	0.947	141.9	-88.0	-0.850	Yes	
J869	50	±21.47	6 x 0.6 MVar Cap	49.4	17.9	0.940	49.1	-33.1	-0.829	Yes	

Project	Pmax (MW)	Reactive Power Capability (MVar)	Proposed VAR Compensation	VAR Injection			VAR Absorption			Meet FERC Order 827 Requirement?	Add'l VAR Needed (MVar)
				P (MW)	Q (MVar)	Lagging p.f (pu)	P (MW)	Q (MVar)	Leading p.f. (pu)		
J873	200	41 -58.3	4 x 15 MVar Cap	191.2	60.4	0.953	186.7	-146.3	-0.787	No	3
J874	150	±72.6	N/A	147.6	42.8	0.960	145.6	-129.1	-0.748	No	5.7
J876	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
J877	250	±121	N/A	247.2	79.5	0.952	245.7	-187.9	-0.794	No	1.8
J880	150	±50	2 x 15 MVar Cap	149.1	68.2	0.910	148.7	-83.8	-0.871	Yes	
J885	64	±31	N/A	63.2	20.7	0.950	62.9	-45.4	-0.811	Yes	
J889	150	±50	2 x 15 MVar Cap	148.5	66.6	0.912	147.8	-86.1	-0.864	Yes	
J891	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
J897	190	±93.16	2 x 8 MVar Cap	188.0	69.6	0.938	185.8	-182.8	-0.713	Yes	
J898	100	±19.72	1 x 8 MVar Cap	99.1	9.9	0.995	98.8	-43.2	-0.916	No	22.7
J901	200	±35.12	2 x 8 MVar Cap	198.1	0.6	1.000	197.4	-103.9	-0.885	No	64.5
J905	41	±13	2 x 3 MVar Cap	40.1	13.7	0.946	40.2	-23.3	-0.865	Yes	
J916	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
J926	102.5	±48.907	1 x 8 MVar Cap	101.3	43	0.921	101.3	-76.2	-0.800	Yes	
J927	200	±66	3 x 8 MVar Cap	196.7	56.9	0.961	195.0	-119.7	-0.852	No	7.8
J929	25	±0	3 x 4 MVar Cap 2 x 4 MVar Reactor	24.9	11.3	0.911	24.9	-11.3	-0.911	Yes	
J933	211	±67	3 x 8 MVar Cap	210.0	69.9	0.949	209.7	-102.1	-0.899	Yes	
J946	210	±69.4	2 x 20 MVar Cap	200	74.4	0.865	124.9	-68.1	-0.878	Yes	

3. Model Development and Study Assumptions

3.1. Base Case Models

The origin of the DPP 2017 August West models is based on the MTEP 18 series models with the Bench Cases including all prior-queued projects and their associated network upgrades known through the DPP February West Phase 2 analysis, while the Study Cases contain all of the interconnection requests in DPP 2017 August West Phase 1 as well as the base case upgrade of Big Stone South – Alexandria 345 kV, in addition to the Bench Cases. Projects J876 and J891 were withdrawn since they did not submit their model review forms in accordance with Attachment X of the MISO tariff. These projects were removed from the study cases.

Prior queued network upgrades are documented in the following System Impact Study reports

MISO DPP 2017 February West Area Phase 2 Study section 4.5

https://cdn.misoenergy.org/GI-DPP_2017-FEB-West_Phase2_System_Impact_Report_Public366110.pdf



MISO DPP 2016 August West Area Phase 3 Study section 4.3

https://cdn.misoenergy.org/GI-DPP-2016-AUG-Phase3_System_Impact_Report_PUBLIC328119.pdf

MISO DPP 2016 August West Area Phase 3 Study section 2.8

https://cdn.misoenergy.org/GI_DPP_2016_FEB_West_Phase3_SIS_Public_Final269535.pdf

- Bench Cases
 - DPP_AUG17_West_2023SH90_Bench_FINAL_190725
 - DPP_AUG17_West_2023SUM_Bench_FINAL_190725
- Study Cases
 - DPP_AUG17_West_2023SH90_Study_FINAL_190725
 - DPP_AUG17_West_2023SUM_Study_FINAL_190725

3.2. Monitored Elements

Under NERC category P0 conditions (system intact) branches were monitored for loading above the normal rating (PSS®E Rating A), and for NERC category P1-P7 conditions branches were monitored for emergency rating (PSS®E Rating B). Voltage limits were specified for system intact and contingent conditions as per applicable Transmission Owner Planning Criteria.

3.3. Contingencies

The following contingencies were considered in the steady state analysis:

- 1) NERC Category P0 (system intact -- no contingencies)
- 2) NERC Category P1 contingencies
 - a. Single element outages, at buses with a nominal voltage of 68 kV and above
 - b. Multiple element NERC Category P1 contingencies
- 3) NERC Category P2-P7 contingencies
- 4) For all the contingencies and post-disturbance analyses, cases were solved with transformer tap adjustment enabled, area interchange adjustment disabled, phase shifter adjustment disabled (fixed) and switched shunt adjustment enabled.

3.4. Study Methodology

Non-linear (AC) contingency analysis was performed on the benchmark and study cases, and the incremental impact of the DPP 2017 August West generating facilities was evaluated by comparing the steady state performance of the transmission system in the Bench and Study Cases. Analyses used PSS®E version 33.11.0 and TARA version 1801d.

3.5. Performance Criteria

A branch is considered a thermal constraint if the following conditions are met:

- 1) The generator has a larger than twenty percent (20%) sensitivity factor on the overloaded facilities under post-contingent condition (see NERC TPL) or five percent (5%) sensitivity factor under system-intact condition, or
- 2) The overloaded facility or the overload-causing contingency is at generator's outlet, or
- 3) The megawatt impact due to the generator is greater than or equal to twenty percent (20%) of the applicable rating (normal or emergency) of the overloaded facility, or
- 4) For any other constrained facility, where none of the Study Generators meet one of the above criteria, however, the cumulative MW impact of the group of study generators is greater than twenty percent (20%) of the rating of the facility, then only those study generators whose individual MW impact is greater than five percent (5%) of the rating of the facility and has DF greater than five percent (5%) will be responsible for mitigating the cumulative MW impact constraint, or
- 5) Impacts on Affected Systems would be classified as Injection constraints based on the Affected Systems' criteria, or
- 6) Any other applicable Transmission Owner FERC filed Local Planning Criteria are met.

A bus is considered a voltage constraint if both of the following conditions are met:

- 1) The bus voltage is outside of the applicable normal or emergency limits for the post change case, and
- 2) The change in bus voltage is greater than 0.01 per unit

All generators must mitigate thermal injection constraints and voltage constraints in order to obtain any type of Interconnection Service. Further, all generators requesting Network Resource Interconnection Service (NRIS) must mitigate constraints found by using the Deliverability algorithm, to meet the system performance criteria for NERC category P1 events, if DFAX due to the study generator is equal to or greater than 5%.

4. Backbone Network Upgrade Analysis

An analysis was performed to determine the need for any backbone network upgrades for this cycle. Significant low voltages across northwest Minnesota and eastern North Dakota were observed for the 345 kV P1 contingencies. Many of the voltages were observed to be in the mid-70s to low 80s percent range for this single outage. Solutions considered included the Big Stone South – Alexandria 345 kV line and a Bison – Big Stone South 345 kV line.

The Bison – Big Stone South 345 kV is a new 345 kV connection that was assumed to be approximately 65 miles with a planning level estimate cost of \$135 million. The Bison-Big Stone South was ineffective in providing acceptable voltage constraint relief under P1 contingency condition. The Big Stone South – Bison 345 kV line only resulted in a system intact flow of approximately 200 MWs. Under the critical contingency, the flow on the proposed mitigation drops to approximately 40 MWs. This proposed mitigation provides the necessary flow relief and consequently no voltage relief.

The Big Stone South – Alexandria 345 kV line is a network upgrades that has been used in past DPP studies to address significant issues in the study cases. This line is assumed to be approximately 95 miles in length with a planning level estimate cost of \$197.4 million. The system intact flow on this line is over 600 MWs. Under the critical contingency, the flows increase to 800 MWs. More importantly, system voltages in the northwest Minnesota and eastern North Dakota are significantly better post contingent condition than the case without Big Stone South – Alexandria 345 kV. Based on the effectiveness of the Big Stone South – Alexandria 345 kV in addressing significant amount of voltage violations, this network upgrades were selected to be included in the study case as a backbone network upgrade.

The upgrade was further refined to terminate at the Twin Brooks substation in lieu of the Big Stone South substation to address a voltage collapse condition for the loss of another 345kV element as a P1 contingency. This change is expected to add approximately 30 miles compared to the Big Stone South – Alexandria 345 kV line to bring the line mileage total to 125 miles. This also leads to a planning level cost estimate of \$242.4 million. This was identified as lower cost than Big Stone South – Alexandria alone with a second Twin Brooks – Big Stone South 345 kV circuit with an estimated cost of \$54.5 million as identified in the 2017 February West Phase I report. The performance of Twin Brooks – Alexandria 345 kV is comparable to Big Stone South – Alexandria with respect to addressing voltage issues in northwest Minnesota and eastern North Dakota while also addressing the voltage collapse issue.

5. Thermal Analysis

The thermal analysis results for 2017 August West group show generator projects causing constraints. The details pertaining to the thermal analysis can be found in Appendix A – .

6. Voltage Analysis

The voltage analysis results for summer peak models show that the no study generators cause any voltage constraints. The shoulder peak analysis does indicate that three contingencies result in a voltage collapse condition that will require mitigation by the August 2017 study group. The details pertaining to the voltage analysis can be found in Appendix A – .

7. Stability Analysis

Transient stability analysis will be performed in the 2017 August West Phase 2 study. A voltage stability analysis was performed for the Minnesota-Wisconsin Export Interface (MWEX). The results of that analysis indicate that the August 2017 Study group does not aggravate the interface and no network upgrades were therefore identified. The full analysis is included in appendix B.

8. Short Circuit Analysis

Short circuit analysis will be performed in the 2017 August West Phase 2 study.

9. Affected System Impact Study

Affected System analysis will be performed in the 2017 August West Phase 2 study.

10. Deliverability Analysis

10.1. Introduction

Generator interconnection projects have to pass Generator Deliverability Study to be granted NRIS. If the generator is deemed not fully deliverable, the customer can choose either to change the project to an Energy Resource (ER) project or to proceed with the system upgrades that will make the generator fully deliverable. Generator Deliverability Study ensures that the Network Resources, on an aggregate basis, can meet the MISO aggregate load requirements during system peak condition without getting bottled up. The wind generators are tested at 100% of their maximum output level which then can be used to meet Resource Adequacy obligations, under Module E, of the MISO Transmission and Energy Market Tariff (TEMT).

MISO Generator Deliverability Study whitepaper describing the algorithm can be found in BPM 015 – Generation Interconnection, Appendix C.

10.2. Determining the MW Restriction

If one facility is overloaded based on the assessed “severe yet credible dispatch” scenario described in the study methodology, and the generator under study has a DF greater than 5%, part or all of its output is not deliverable. The restricted MW is calculated as following:

$$(\text{MW restricted}) = (\text{worst loading} - \text{MW rating}) / (\text{generator sensitivity factor})$$

If the result is larger than the maximum output of the generator, 100% of this generator’s output is not deliverable.

10.3. Deliverability Study Results

The deliverability analysis assumes all ERS upgrades as those upgrades are required for the study group to interconnection to the transmission system. With all the MISO analysis ERS upgrades assumed as well as the prior queued network upgrades, the deliverability analysis determined no new constraints for projects that are requesting NRIS service.

10.3.1. G517

This generator did not request NRIS service, so deliverability was not performed for this interconnection request.

10.3.2. J447

This generator is determined to be fully deliverable for 190 MW, contingent upon the ERS system upgrades and assumed prior queued network upgrades.

10.3.3. J545

This generator is determined to be fully deliverable for 110 MW, contingent upon the ERS system upgrades and assumed prior queued network upgrades.

10.3.4. J580

This generator did not request NRS service, so deliverability was not performed for this interconnection request

10.3.5. J628

This generator is determined to be fully deliverable for 400 MW, contingent upon the ERS system upgrades and assumed prior queued network upgrades.

10.3.6. J705

This generator is determined to be fully deliverable for 100 MW, contingent upon the ERS system upgrades and assumed prior queued network upgrades.

10.3.7. J706

This generator is determined to be fully deliverable for 100 MW, contingent upon the ERS system upgrades and assumed prior queued network upgrades.

10.3.8. J713

This generator is determined to be fully deliverable for 300 MW, contingent upon the ERS system upgrades and assumed prior queued network upgrades.

10.3.9. J720

This generator is determined to be fully deliverable for 200 MW, contingent upon the ERS system upgrades and assumed prior queued network upgrades.

10.3.10. J722

This generator is determined to be fully deliverable for 200 MW, contingent upon the ERS system upgrades and assumed prior queued network upgrades.

10.3.11. J743

This generator is determined to be fully deliverable for 200 MW, contingent upon the ERS system upgrades and assumed prior queued network upgrades.

10.3.12. J785

This generator is determined to be fully deliverable for 105 MW, contingent upon the ERS system upgrades and assumed prior queued network upgrades.

10.3.13. J801

This generator is determined to be fully deliverable for 74 MW, contingent upon the ERS system upgrades and assumed prior queued network upgrades.

10.3.14. J803

This generator did not request NRS service, so deliverability was not performed for this interconnection request.

10.3.15. J816

This generator is determined to be fully deliverable for 110 MW, contingent upon the ERS system upgrades and assumed prior queued network upgrades.

10.3.16. J836

This generator i did not request NRS service, so deliverability was not performed for this interconnection request.

10.3.17. J840

This generator is determined to be fully deliverable for 150 MW, contingent upon the ERIIS system upgrades and assumed prior queued network upgrades.

10.3.18. J869

This generator is determined to be fully deliverable for 49.98 MW, contingent upon the ERIIS system upgrades and assumed prior queued network upgrades.

10.3.19. J873

This generator is determined to be fully deliverable for 200 MW, contingent upon the ERIIS system upgrades and assumed prior queued network upgrades.

10.3.20. J874

This generator is determined to be fully deliverable for 150 MW, contingent upon the ERIIS system upgrades and assumed prior queued network upgrades.

10.3.21. J877

This generator is determined to be fully deliverable for 250 MW, contingent upon the ERIIS system upgrades and assumed prior queued network upgrades.

10.3.22. J880

This generator is determined to be fully deliverable for 150 MW, contingent upon the ERIIS system upgrades and assumed prior queued network upgrades.

10.3.23. J885

This generator is determined to be fully deliverable for 64 MW, contingent upon the ERIIS system upgrades and assumed prior queued network upgrades.

10.3.24. J889

This generator is determined to be fully deliverable for 150 MW, contingent upon the ERIIS system upgrades and assumed prior queued network upgrades.

10.3.25. J901

This generator is determined to be fully deliverable for 200 MW, contingent upon the ERIIS system upgrades and assumed prior queued network upgrades.

10.3.26. J916

This generator is determined to be fully deliverable for 40 MW, contingent upon the ERIIS system upgrades and assumed prior queued network upgrades.

10.3.27. J926

This generator is determined to be fully deliverable for 101.28 MW, contingent upon the ERIIS system upgrades and assumed prior queued network upgrades.

10.3.28. J927

This generator is determined to be fully deliverable for 100 MW, contingent upon the ERIIS system upgrades and assumed prior queued network upgrades.

10.3.29. J929

This generator is determined to be fully deliverable for 25 MW, contingent upon the ERIIS system upgrades and assumed prior queued network upgrades.

10.3.30. J933

This generator is determined to be fully deliverable for 200 MW, contingent upon the ERIIS system upgrades and assumed prior queued network upgrades.

10.3.31. J946

This generator is determined to be fully deliverable for 200 MW, contingent upon the ERIS system upgrades and assumed prior queued network upgrades.

11. Shared Network Upgrades Analysis

Shared Network Upgrade (SNU) Analysis tests for Network Upgrades driven by higher queued interconnection projects was performed for this System Impact Study. SNUs were identified for DPP 2017 August West Area Projects.

The maximum MW impacts and Shared Network Upgrade (SNU) cost allocations appear in Table 6.

Table 6: Maximum MW Impact and SNU Cost Allocations

Network Upgrades	Project Study Cycle	Projects sharing cost	MW Contribution	Total NU Cost (\$)	Cost Responsibility (\$)
620362 OAKES 4 620363 FORMAN 4 1	DPP-2015-AUG	J436	11.64	\$300,000	\$32,733
	DPP-2015-AUG	J437	11.64		\$32,733
	DPP-2016-FEB	J488	11.72		\$32,958
	Merricourt	G359R	40.43		\$113,695
	DPP-2017-AUG	J580	26.03		\$73,200
	DPP-2017-AUG	J929	5.22		\$14,679
620362 OAKES 4 620363 FORMAN 4 1	DPP-2016-FEB	J488	11.72	\$950,000	\$133,501
	Merricourt	G359R	40.43		\$460,533
	DPP-2017-AUG	J580	26.03		\$296,504
	DPP-2017-AUG	J929	5.22		\$59,460
620362 OAKES 4 620363 FORMAN 4 1	Merricourt	G359R	40.43	\$112,750	\$63,594
	DPP-2017-AUG	J580	26.03		\$40,944
	DPP-2017-AUG	J929	5.22		\$8,210
620362 OAKES 4 620363 FORMAN 4 1	DPP-2016-AUG	J302	15.7	\$4,861,933	\$1,286,140
	DPP-2016-AUG	J503	15.4		\$1,261,156
	DPP-2017-AUG	J580	23.03		\$1,886,610
	DPP-2017-AUG	J929	5.22		\$427,621
620314 BIGSTON4 655465 BLAIR-ER4 1	DPP-2016-FEB	J488	22.64	\$150,183	\$34,855
	DPP-2016-FEB	J493	7.63		\$11,746
	DPP-2016-FEB	J526	19.47		\$29,975
	DPP-2017-AUG	J722	47.81		\$73,605
620314 BIGSTON4 620325 BROWNSV4 1	DPP-2016-FEB	J488	8.97	\$285,086	\$48,068
	DPP-2016-FEB	J493	6.82		\$36,546
	DPP-2016-FEB	J526	14.89		\$79,791
	DPP-2017-AUG	J722	22.52		\$120,679

12. Cost Allocation

The cost allocation of Network Upgrades for the study group reflects responsibilities for mitigating system impacts based on Interconnection Customer-elected level of Interconnection service as of the draft System Impact Study report date.

12.1. Cost Assumptions for Network Upgrades

The cost estimate for each network upgrade identified in System Impact Study was provided by the corresponding transmission owning company.

12.2. Cost Allocation Methodology

The costs of Network Upgrades (NU) for a set of generation projects (one or more sub-groups or entire group with identified NU) are allocated based on the MW impact from each project on the constrained facilities in the Study Case.

Cost Allocation Methodology for Thermal Constraints

1. With all Study Group generation projects dispatched in the Study Case, all thermal constraints are identified.
2. Distribution factor from each project on each constraint is obtained.
3. For each thermal constraint, the maximum MW contribution (increasing flow) from each project is then calculated in the Post Case without any network upgrades.
4. For each thermal constraint, the cost estimates for one or a subset of NU are provided by the corresponding Transmission Owner.
5. Then the cost of each NU is allocated based on the pro rata share of the MW contribution from each project on the constraints mitigated or partly mitigated by this NU. The methodology to determine the cost allocation of one NU is:

$$\text{Cost of NU} = \frac{\text{Project A cost portion of NU} \times \text{Max(Proj. A MW contribution on constraint)}}{\sum_i \text{Max(Proj. i MW contrution on constraint)}}$$

6. The total NU costs for each project are calculated if more than one NU is required.

Cost Allocation Methodology for Voltage Constraints

Cost allocation of voltage constraint driven network upgrades will be determined by the pro rata share of the voltage impact each project has on the most constrained bus under the most constraining contingency. The voltage impact of each project will be calculated by locking all voltage regulating equipment in the model and then backing out each GI project one at a time to identify each project's impact to the constraint. In severe instances of voltage collapse where projects cannot be backed out one at a time, they will be added one at a time to determine their impact to the constraint.

As the number and types of constraints increases, mitigating the constraints individually may result in higher overall costs. In instances when mitigation(s) resolve multiple types of constraints (such as thermal + voltage or thermal + voltage + transient stability) the cost is allocated based off the ratio share of the total cost of the independent mitigations in order to equitably allocate the cost to all parties contributing to constraints. In summary, only the lowest cost mitigation option will be constructed, but for cost allocation purposes the independent mitigations are required.



Appendix A – Thermal Analysis Results

Thermal Constraints

Appendix B – Voltage Constraints

Voltage Constraints

Appendix C – MWEX Voltage Stability Report



**MWEX Voltage Stability Study Report
Prepared For
August 2017 Phase 1 MISO West Area DPP Study**

**August 5, 2019
American Transmission Company, LLC**

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Summary

Voltage stability analysis is required to determine if the initial conditions of the Definitive Planning Phase (DPP) system models under study are in a stable state as defined by Power-Voltage (PV) curves of the Minnesota Wisconsin Export Interface (MWEX) for the worst contingency. The worst contingency was identified as the P7.1 outage of Superior – Stone Lake 345 kV and Stinson – Hawthorne – Gordon 161 kV lines based on a full steady state voltage stability analysis of more than 100 contingencies. MWEX is the summation of the flows on the AS King – Eau Claire 345 kV line measured at AS King and the Arrowhead 230 kV phase shifting transformer (also referred to as phase angle regulator or “PAR”) measured at the Minnesota Power 230 kV side of the Arrowhead substation. The voltage is measured at the Arrowhead 230 kV bus. Also, normal and emergency voltage limits will be respected in addition to steady state voltage stability limits. The 2023 shoulder load (2023SH) cases provided by MISO required a few modifications as discussed in Appendix A to produce “VSAT Ready” cases.

The voltage stability analysis used 2023SH cases to compare the Pre-DPP without Network Upgrades (referred to as “Pre-DPP” hereafter) and the Post-DPP with the Base Case Network Upgrade as provided by MISO (referred to as “Post-DPP + Base Case NU” hereafter). The Base Case NU is the Alexandria – Big Stone 345 kV line.

Also, both the Pre-DPP and Post-DPP + Base Case NU cases contain the Network Upgrades required for the previous West Area generation interconnection studies including but not limited to the Cardinal – Hickory Creek MVP (Cardinal – Hill Valley – Hickory Creek 345 kV with a 345/138 kV transformer at Hill Valley, MTEP PrjID 3127) and the Webster – Franklin – Morgan Valley 345 kV line.

As shown in Table 7, the Pre-DPP scenario in the 2023SH case is not voltage stable. The Pre-DPP scenario does not converge with the worst contingency and therefore is in a violation of ATC Planning Criteria. The Post-DPP scenario in the 2023SH case is voltage stable but is also in violation of ATC Planning Criteria because the voltage stability margin is less than 10% with the worst contingency. The Post-DPP scenario is not aggravating the criteria violation because of the dispatch of August 2017 DPP requests outside of the West Area, including the J732 request at the ATC Superior 345 kV Substation.

Because the Post-DPP scenario is not aggravating the criteria violations, Network Upgrades related to voltage stability will not be assigned to the Interconnection Customers, based on the assumptions used in this analysis.



MISO DPP 2017 August West Area Phase 1 Local Planning Criteria Study

Prepared for
MISO

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Revision History

Date	Rev.	Description
9/26/2019	A	Draft Report
10/2/2019	B	Final Report

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Executive Summary

In addition to MISO's standard Definitive Planning Phase (DPP) analysis, Local Planning Criteria (LPC) analyses were performed to identify additional constraints per Transmission Owning Companies' LPC. This report presents the LPC study results for Dairyland Power Cooperative (DPC) LPC, Montana-Dakota Utilities Company (MDU) LPC, Otter Tail Power (OTP) LPC, and Great River Energy (GRE) LPC.

1.1 DPC LPC Study

J801 and J898 are to be interconnected in DPC transmission system. DPC determined that J801 and J898 are required for DPC LPC study. The DPC LPC analysis consisted of steady-state contingency analysis for summer shoulder and summer peak conditions.

1.1.1 J801 DPC LPC Study

Additional Network Upgrades required in the J801 DPC LPC study are listed in Table ES-1.

Table ES-1: Additional Network Upgrades for Constraints Identified in J801 DPC LPC Analysis

Constraint	Owner	Worst Loading		Generator	Mitigation	Cost (\$)
		(MVA)	(%)			
Beaver Creek-Harmony 161 kV	DPC	397.8	105.2	J898	Raise some structures	\$2,000,000
Wabaco-Rochester 161 kV ¹	DPC	258.6	116.9	J898	Rebuild line with 795 ACSS conductor	\$11,000,000

Note 1: J898 will assume cost responsibility if the Wabaco-Rochester 161 kV rebuild is no longer an approved MTEP Appendix A project.

1.1.2 J898 DPC LPC Study

Additional Network Upgrades required in the J898 DPC LPC study are listed in Table ES-2.

Table ES-2: Additional Network Upgrades for Constraints Identified in J898 DPC LPC Analysis

Constraint	Owner	Worst Loading		Generator	Mitigation	Cost (\$)
		(MVA)	(%)			
Beaver Creek-Harmony 161 kV	DPC	424.1	112.2	J898	Raise some structures	\$2,000,000
Wabaco-Rochester 161 kV ¹	DPC	330.8	149.6	J898	Rebuild line with 795 ACSS conductor	\$11,000,000
Wabaco-Alma 161 kV	DPC	306.9	105.5	J898	Replace bus and jumpers in Rochester substation	\$200,000

Note 1: J898 will assume cost responsibility if the Wabaco-Rochester 161 kV rebuild is no longer an approved MTEP Appendix A project.

1.1.3 Summary of Additional Network Upgrades Identified in DPC LPC Analysis

In summary, additional Network Upgrades required in the DPC LPC study are listed in Table ES-3.

Table ES-3: Summary of Additional Network Upgrades Required in DPC LPC Analysis

Constraint	Owner	Worst Loading		Mitigation	Cost (\$)
		(MVA)	(%)		
Beaver Creek-Harmony 161 kV	DPC	424.1	112.2	Raise some structures	\$2,000,000
Wabaco-Rochester 161 kV ¹	DPC	330.8	149.6	Rebuild line with 795 ACSS conductor	\$11,000,000
Wabaco-Alma 161 kV	DPC	306.9	105.5	Replace bus and jumpers in Rochester substation	\$200,000

Note 1: J898 will assume cost responsibility if the Wabaco-Rochester 161 kV rebuild is no longer an approved MTEP Appendix A project.

1.2 MDU LPC Study

MDU determined that J580 has more than 20% Distribution Factor (DF) on the outlets of J302, J503 and J741, J929 has more than 20% DF on the outlets of Tatanka, Foxtail, J436, J437, J488, J302, J503, G359 and J741, and J933 has more than 20% DF on the outlets of J436, J437 and J488. Therefore, J580, J929, and J933 are required for OTP LPC study per Section 3.2 of the MDU Local Planning Criteria. The MDU LPC analysis consisted of steady- state contingency analysis for summer shoulder condition.

1.2.1 Summary of Additional Network Upgrades Identified in MDU LPC Analysis

In summary, additional Network Upgrades required in the MDU LPC study are listed in Table ES-4.

Table ES-4: Additional Network Upgrades for Constraints Identified in MDU LPC Analysis

Constraint	Owner	Worst Loading		Generator	Mitigation	Cost (\$)
		(MVA)	(%)			
J580 POI-J302 POI 230 kV	MDU	400.8	104.6	J929	Add breaker at Merricourt	\$1,500,000
J580 POI-Heskett 230 kV	MDU	444.8	116.1	J580,J929	Add breaker at Merricourt	Incl.
Ellendale-Aberdeen Jct 115 kV	MDU	160.0	125.0	J933	Minor Substation upgrades	\$60,000

Constraint	Owner	Worst Loading		Generator	Mitigation	Cost (\$)
		(MVA)	(%)			
	NWPS					
West Bismarck-Bismarck 115 kV	MDU WAPA	211.9	138.5	J580,J713	MDU's rating for line and East Bismarck is 249 MVA. WAPA owns equipment at Bismarck.	\$0
Glenham-Campbell 230 kV	MDU WAPA	301.0	143.3	J580, J628, J713, J743, J880, J897, J929, J933	MDU's rating at Glenham is 351 MVA. WAPA owns equipment at Campbell and line.	\$0
Mandan-Ward 230 kV	MDU BEPC WAPA MPC	485.4	124.2	J580,J929,J933	MDU's rating at Mandan is 956 MVA. MPC owns equipment at Mandan. WAPA owns the line. BEPC owns Ward.	\$0
Fox Tail-Tatanka North 230 kV	MDU	521.3	109.1	J580,J929	Replace breakers	\$2,000,000
Fox Tail-Ellendale MVP 230 kV	MDU	663.9	108.8	J580,J929	Rebuild Line	\$10,500,000
Merricourt-Tatanka North 230 kV	MDU	350.3	102.1	J580,J929	Reconductor Line	\$1,000,000
Merricourt-Ellendale MVP 230 kV	MDU	646.8	106.0	J580,J929	Rebuild Line	\$18,000,000
Ellendale MVP-Oakes 230 kV	MDU OTP	563.9	107.0	J929,J933	MDU's rating at Mandan is 776 MVA. OTP owns equipment at Oakes and the line.	\$0

1.3 OTP LPC Study

J722, J816, and J933 are to be interconnected in OTP transmission system. OTP determined that J722, J816, and J933 are required for OTP LPC study. The OTP LPC analysis consisted of steady-state contingency analysis for summer peak, summer shoulder, and light load no wind (LLNW) conditions.

1.3.1 Summary of Additional Network Upgrades Identified in OTP LPC Analysis

In summary, additional Network Upgrades required in the OTP LPC study are listed in Table ES-5

Table ES-5: Additional Network Upgrades for Constraints Identified in OTP LPC Analysis

Constraint	Worst Loading		Generator	Mitigation	Owner	Cost (\$)
	(MVA)	(%)				
Canby-Granite Falls 115 kV	155.1	137.3	J722,J933	OTP requires 39.22 miles of line rebuild and CT adjustments.	OTP WAPA	\$17,700,000

Constraint	Worst Loading		Generator	Mitigation	Owner	Cost (\$)
	(MVA)	(%)				
Winger-Fertile 115 kV	109.2	107.2	J628	OTP requires additional line clearance mitigations beyond the MISO upgrades.	OTP	\$450,000
Wilton-Bemidji 115 kV	204.6	109.8	J628,J897	OTP requires additional line clearance mitigations beyond the MISO upgrades.	OTP MPC	\$50,000
Bemidji-Helga 115 kV	173.1	108.2	J628,J889,J897	OTP equipment sufficient for flow.	OTP MPC	\$0
Crookston-Fertile 115 kV	112.9	107.0	J628	OTP requires additional line clearance mitigations beyond the MISO upgrades.	OTP	\$250,000
Donalds-Warsaw 115 kV	118.0	101.8	J628	OTP requires relay replacement/adjustments.	OTP MPC	\$50,000
Hensel-Drayton 115 kV	137.4	103.3	J897	OTP equipment sufficient for flow.	OTP MPC	\$0
Hensel-ENIDTP 115 kV	164.8	100.5	J897	OTP equipment sufficient for flow.	OTP MPC	\$0
Ramsey 230-115-12.5 kV xfmr	231.4	165.3	J889,J897	Non-OTP owned equipment	GRE	\$0
Wahpeton-Fergus Falls 230 kV	417.3	117.1	J743,J933,J946	OTP requires clearance mitigations.	OTP MRES	\$1,150,000
Audubon-Hubbard 230 kV	380.7	100.1	J743,J946	OTP requires additional line clearance mitigations beyond the MISO upgrades.	OTP GRE XEL MP	\$250,000
Oakes-Forman 230 kV	530.1	100.8	J929,J933	Uprates per August 2016 cycle will be sufficient for these flows	OTP	\$0
Oakes-Ellendale 230 kV	536.6	101.8	J929,J933	Uprates per August 2016 cycle will be sufficient for these flows	OTP MDU	\$0
Voltage constraints				Install a breaker at Bison 345 kV (bus #: 601067)	XEL	\$1,500,000
Voltage constraints				1×60 Mvar switched cap bank at Silver Lake 230 kV (615637)	GRE	\$2,500,000
Voltage constraints				1×75 Mvar switched cap bank at Jamestown 345 kV (bus #: 620369)	OTP	\$4,000,000
Voltage constraints				Additional 3×25 Mvar switched cap bank at Wahpeton 230 kV (bus #: 620329) (total of 5×25 Mvar)	OTP	\$3,000,000
Voltage constraints				1×40 Mvar switched cap bank at Audubon 230 kV (bus #: 620336)	MRES	\$1,500,000
Voltage constraints				2×20 Mvar switched cap bank at Perham SE 115 kV (bus #: 620177)	OTP	\$2,000,000
Voltage constraints				1×20 Mvar switched cap bank at Bagley 115 kV (bus #: 620239)	OTP	\$1,000,000

Constraint	Worst Loading		Generator	Mitigation	Owner	Cost (\$)
	(MVA)	(%)				
Voltage constraints				Additional 1×16.2 Mvar switched cap bank at Alexandria 115 kV (bus #: 620222) (total of 2×16.2 Mvar)	OTP	\$1,000,000

1.4 GRE LPC Study

GRE determined that the GRE LPC should be applied to J628, J705, J706, J713, J869, J880, J889, J897, and J901 because these projects utilize GRE-owned transmission to access to the MISO network. The GRE LPC analysis consisted of steady-state contingency analysis for summer shoulder condition.

1.4.1 J869 GRE LPC Study

No additional Network Upgrades are required in the J869 GRE LPC study. All the thermal constraints are mitigated by the MISO NUs listed in Table ES-6.

Table ES-6: MISO Network Upgrades for Constraints Identified in J869 GRE LPC Analysis

Constraint	Owner	Mitigation ¹	Cost (\$)	Comments
Dotson Corners-City of Springfield 69 kV	GRE ITCM	Uprate the line	\$60,000	MISO NUs
Cobden Tap-Brookville 69 kV	GRE ITCM	Rebuild the line	\$2,570,000	MISO NUs
Cobden Tap-City of Springfield 69 kV	GRE ITCM	Rebuild the line	\$396,000	MISO NUs
Sundown-Brookville 69 kV	GRE	Rebuild the line	\$2,700,000	MISO NUs
Jefferson Tap-N Storden 69 kV	GRE	Rebuild the line	\$2,900,000	MISO NUs

Note 1: All these mitigations are required in MISO analysis. No additional mitigations are required in J869 GRE LPC study.

1.4.2 J901 GRE LPC Study

Additional Network Upgrades required in the J901 GRE LPC study are listed in Table ES-7.

Table ES-7: Additional Network Upgrades for Constraints Identified in J901 GRE LPC Analysis

Constraint	Owner	Mitigation	Cost (\$)	Comments
Dotson Corners-City of Springfield 69 kV	GRE ITCM	Uprate the line	\$60,000	MISO NUs ¹
Cobden Tap-Brookville 69 kV	GRE ITCM	Rebuild the line	\$2,570,000	MISO NUs ¹

Constraint	Owner	Mitigation	Cost (\$)	Comments
Cobden Tap-City of Springfield 69 kV	GRE ITCM	Rebuild the line	\$396,000	MISO NUS ¹
Sundown-Brookville 69 kV	GRE	Rebuild the line	\$2,700,000	MISO NUS ¹
Voltage constraints	GRE	1×60 Mvar switched cap bank at Silver Lake 230 kV (615637)	\$2,500,000	J901 GRE LPC
Voltage constraints	GRE	1×60 Mvar switched cap bank at Panther 230 kV (615529)	\$2,500,000	J901 GRE LPC

Note 1: All these mitigations are required in MISO analysis. No additional mitigations are required in J901 GRE LPC study.

1.4.3 CCS GRE LPC Study

Additional Network Upgrades required in the CCS GRE LPC study are listed in Table ES-8.

Table ES-8: Additional Network Upgrades for Constraints Identified in CCS GRE LPC Analysis

Constraint	Mitigation	Owner	Cost (\$)	Comments
Ramsey 230-115-12.5 kV xfmr	Replace xfmr	GRE	\$5,000,000	CCS GRE LPC study
Hubbard-Audubon 230 kV	GRE facility is not a limitation. \$0	GRE OTP XEL MP	\$0	CCS GRE LPC study
McHenry-Voltair 115 kV	GRE facility is not a limitation. \$0	GRE XEL BEPC	\$0	CCS GRE LPC study
Silver Lake-Henning 230 kV	GRE facility is not a limitation. \$0	GRE MRES OTP	\$0	CCS GRE LPC study
Silver Lake-Fergus Falls 230 kV	GRE facility is not a limitation. \$0	GRE MRES OTP	\$0	CCS GRE LPC study
Dotson Corners-City of Springfield 69 kV	Update the line	GRE ITCM	\$60,000	MISO NUS ¹
Cobden Tap-Brookville 69 kV	Rebuild the line	GRE ITCM	\$2,570,000	MISO NUS ¹
Cobden Tap-City of Springfield 69 kV	Rebuild the line	GRE ITCM	\$396,000	MISO NUS ¹
Sundown-Brookville 69 kV	Rebuild the line	GRE	\$2,700,000	MISO NUS ¹
Jefferson Tap-N Storden 69 kV	Rebuild the line	GRE	\$2,900,000	MISO NUS ¹

Constraint	Mitigation	Owner	Cost (\$)	Comments
Voltage collapse and voltage constraints in benchmark case	Build Center-Ellendale 345 kV line (200 miles)	GRE MPC MDU	\$406,750,000	CCS GRE LPC Benchmark ²
Voltage collapse and voltage constraints in benchmark case	2×75 Mvar switched cap bank at Jamestown 345 kV (620369)	OTP	\$8,000,000	CCS GRE LPC Benchmark ²
Voltage collapse and voltage constraints in study case	1×60 Mvar switched cap bank at Silver Lake 230 kV (615637)	GRE	\$2,500,000	J901 GRE LPC ³
Voltage collapse and voltage constraints in study case	1×60 Mvar switched cap bank at Panther 230 kV (615529)	GRE	\$2,500,000	J901 GRE LPC ³
Voltage collapse and voltage constraints in study case	Additional 3×25 Mvar switched cap bank at Wahpeton 230 kV (bus #: 620329) (total of 5×25 Mvar)	OTP	\$3,000,000	OTP LPC study ⁴
Voltage collapse and voltage constraints in study case	1×40 Mvar switched cap bank at Audubon 230 kV (bus #: 620336)	MRES	\$1,500,000	OTP LPC study ⁴
Voltage collapse and voltage constraints in study case	2×20 Mvar switched cap bank at Perham SE 115 kV (bus #: 620177)	OTP	\$2,000,000	OTP LPC study ⁴
Voltage collapse and voltage constraints in study case	Build single circuit 345 kV line J628-Bison (90 miles)	GRE	\$198,000,000	CCS GRE LPC study
Voltage collapse and voltage constraints in study case	Install 345-230 kV, 448 MVA xfmr at J628 POI	GRE	\$9,200,000	CCS GRE LPC study
Voltage collapse and voltage constraints in study case	Build 345 kV substation at J628 POI. Move J628 POI to 345 kV bus	GRE	\$4,150,000	CCS GRE LPC study
Voltage collapse and voltage constraints in study case	±75 Mvar SVC at Jamestown 345 kV (620369)	OTP	\$20,000,000	CCS GRE LPC study
Voltage collapse and voltage constraints in study case	2×75 Mvar switched cap bank at River Road 345 kV (615664)	GRE	\$5,000,000	CCS GRE LPC study
Voltage collapse and voltage constraints in study case	additional 1×40 Mvar switched cap bank at Audubon 230 kV (bus #: 620336) (total of 2×40 Mvar)	MRES	\$1,500,000	CCS GRE LPC study

Note 1: All these mitigations are required in MISO analysis. No additional mitigations are required in CCS GRE LPC study.

Note 2: These NUs are required for CCS GRE LPC benchmark case. Study projects in the DPP 2017 August cycle are not responsible for these Network Upgrades.

Note 3: These NUs are also required in J901 GRE LPC.

Note 4: These NUs are also required in OTP LPC.

1.4.4 Summary of Additional Network Upgrades Identified in GRE LPC Analysis

In summary, additional Network Upgrades required in the GRE LPC study are listed in Table ES-9.

**Table ES-9: Summary of Additional Network Upgrades
Required in GRE LPC Analysis**

Constraint	Generator	Mitigation	Owner	Cost (\$)	Comments
Ramsey 230-115-12.5 kV xfmr	J889	Replace xfmr	GRE	\$5,000,000	CCS GRE LPC study ⁵
Hubbard-Audubon 230 kV	J743,J946	GRE facility is not a limitation. \$0	GRE OTP XEL MP	\$0	CCS GRE LPC study ⁵
McHenry-Voltair 115 kV	J880	GRE facility is not a limitation. \$0	GRE XEL BEPC	\$0	CCS GRE LPC study ⁵
Silver Lake-Henning 230 kV	J743,J946	GRE facility is not a limitation. \$0	GRE MRES OTP	\$0	CCS GRE LPC study ⁵
Silver Lake-Fergus Falls 230 kV	J743,J946	GRE facility is not a limitation. \$0	GRE MRES OTP	\$0	CCS GRE LPC study ⁵
Dotson Corners-City of Springfield 69 kV	G517	Uprate the line	GRE ITCM	\$60,000	MISO NUs ¹
Cobden Tap-Brookville 69 kV	G517	Rebuild the line	GRE ITCM	\$2,570,000	MISO NUs ¹
Cobden Tap-City of Springfield 69 kV	G517	Rebuild the line	GRE ITCM	\$396,000	MISO NUs ¹
Sundown-Brookville 69 kV	G517	Rebuild the line	GRE	\$2,700,000	MISO NUs ¹
Jefferson Tap-N Storden 69 kV	G517	Rebuild the line	GRE	\$2,900,000	MISO NUs ¹
Voltage constraints		1×60 Mvar switched cap bank at Silver Lake 230 kV (615637)	GRE	\$2,500,000	J901 GRE LPC ²
Voltage constraints		1×60 Mvar switched cap bank at Panther 230 kV (615529)	GRE	\$2,500,000	J901 GRE LPC ²
Voltage collapse and voltage constraints in benchmark case		Build Center-Ellendale 345 kV line (200 miles)	GRE MPC MDU	\$406,750,000	CCS GRE LPC Benchmark ³
Voltage collapse and voltage constraints in benchmark case		2×75 Mvar switched cap bank at Jamestown 345 kV (620369)	OTP	\$8,000,000	CCS GRE LPC Benchmark ³
Voltage collapse and voltage constraints in study case		Additional 3×25 Mvar switched cap bank at Wahpeton 230 kV (bus #: 620329) (total of 5×25 Mvar)	OTP	\$3,000,000	OTP LPC study ⁴
Voltage collapse and voltage constraints in study case		1×40 Mvar switched cap bank at Audubon 230 kV (bus #: 620336)	MRES	\$1,500,000	OTP LPC study ⁴
Voltage collapse and voltage		2×20 Mvar switched cap bank at Perham SE	OTP	\$2,000,000	OTP LPC study ⁴

Constraint	Generator	Mitigation	Owner	Cost (\$)	Comments
constraints in study case		115 kV (bus #: 620177)			
Voltage collapse and voltage constraints in study case		Build single circuit 345 kV line J628-Bison (90 miles)	GRE	\$198,000,000	CCS GRE LPC study ⁵
Voltage collapse and voltage constraints in study case		Install 345-230 kV, 448 MVA xfmr at J628 POI	GRE	\$9,200,000	CCS GRE LPC study ⁵
Voltage collapse and voltage constraints in study case		Build 345 kV substation at J628 POI. Move J628 POI to 345 kV bus	GRE	\$4,150,000	CCS GRE LPC study ⁵
Voltage collapse and voltage constraints in study case		±75 Mvar SVC at Jamestown 345 kV (620369)	OTP	\$20,000,000	CCS GRE LPC study ⁵
Voltage collapse and voltage constraints in study case		2×75 Mvar switched cap bank at River Road 345 kV (615664)	GRE	\$5,000,000	CCS GRE LPC study ⁵
Voltage collapse and voltage constraints in study case		additional 1×40 Mvar switched cap bank at Audubon 230 kV (bus #: 620336) (total of 2×40 Mvar)	MRES	\$1,500,000	CCS GRE LPC study ⁵

Note 1: All these mitigations are required in MISO analysis. No additional mitigations are required in GRE LPC study.

Note 2: These NUs are required in both J901 and CCS GRE LPC.

Note 3: These NUs are required for CCS GRE LPC benchmark case. Study projects in the DPP 2017 August cycle are not responsible for these Network Upgrades.

Note 4: These NUs are required in both OTP LPC and CCS GRE LPC.

Note 5: These NUs are required in CCS GRE LPC.